

Research Papers

Numerical investigations of temperature distribution in refrigerated vehicles with flow-guided phase change cold storage materials

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ABSTRACT

Maintaining a stable internal temperature inside the refrigerated containers is crucial for ensuring the safe transportation of perishable goods, as temperature fluctuations can affect food safety and quality. These fluctuations often arise due to external climatic condition and the operational demands in the containers. This study aims to address these challenges by analyzing and optimizing the thermal behavior of a refrigerated vehicle container using ANSYS Fluent 2021 R1. The research modifies the conventional polyurethane-insulated design by incorporating phase-change materials (PCMs) to enhance temperature stability. Paraffin-based PCM plates with melting temperatures of 3 °C, 5 °C, and 8 °C are strategically installed on the ceiling and side walls. The study evaluates their effectiveness during refrigerated system downtimes and door openings. Results indicate that PCM plates reduce air temperature fluctuations significantly, with the combined installation on the side walls and roof achieving a reduction of 5.35 °C. Notably, RT8HC plates on the side walls demonstrated superior solidification efficiency of 75.4 % higher than RT3HC plates.

Nomenclature

C_p	Specific heat capacity ($J\ kg^{-1}\ K^{-1}$)
g	Gravitational acceleration ($m\ s^{-2}$)
h	Convection coefficient ($W\ m^{-2}\ K^{-1}$)
k	Turbulent kinetic energy ($m^2\ s^{-2}$)
L_f	Latent heat of fusion (kJ/kg)
p	Pressure
Q_r	Recovery energy (J)
S_e	Energy sources or sinks
t	Time (s)
u	Velocity (m/s)
v	Volume (m^3)

(continued)

PCM	Phase change material
PCCSU	phase-change cold storage unit
RANS	Reynolds averaged Navier Stokes
RMSE	Root mean square error
RT	Rubitherm
SS	Stainless steel AISI 304
SST	Shear stress transport
XPS	Extruded polystyrene

Abbreviations

CFD	Computational fluid dynamic
CTES	Cold thermal energy storage
E	Energy
DOF	Degrees of freedom
GHG	Greenhouse gas
H	Enthalpy
PU	Polyurethane

(continued on next column)

Greek symbols

ρ	Density ($kg\cdot m^{-3}$)
ε	Turbulent dissipation rate ($m^2\ s^{-3}$)
μ	Dynamic viscosity
λ	Thermal conductivity ($W\cdot m^{-1}\cdot K^{-1}$)
φ	Phase fraction
ω	Specific turbulence dissipation rate (s^{-1})
η	Efficiency (%)

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Suffix

l	Liquid
f	Fluid
s	Solid
fp	Flash point
o	Open door
w	Wall

1. Introduction

Refrigerated vehicles are a vital link between the food processing and the consumer. About 30 % of global food production is lost due to inadequate handling the cold chain transportation. Worldwide there are about 4 million refrigerated road vehicles [1]. It is predicted that by 2030, global road freight transport will have grown by 2.5 % a year thereby resulting in high energy consumption to maintain the optimum cooling load. Energy required for transportation of frozen foods is more in summer due to high-intensity solar irradiation. The intense summer sunlight increases the heat load, necessitating more energy to maintain the desired internal temperature of the refrigerated container. In contrast, heat load in winter is lower, leading to relatively stable energy consumption. Furthermore, during urban deliveries, the repeated opening and closing of doors consumes more energy as warm air is entering the container. The energy consumption of refrigerated vehicles is significantly influenced by the need to maintain low temperatures inside the container. This is particularly crucial during the intra-city transportation of perishable goods. Refrigeration systems account for approximately 60–70 % of these containers' total electrical energy consumption [2–5]. This high percentage of energy consumption is due to the continuous operation of the refrigeration units needed to maintain the desired temperature, especially during periods when the container doors are frequently opened and closed, such as during loading and unloading activities. Research on refrigerated vehicles has advanced significantly to reduce energy consumption. The integration of thermal energy storage units, such as PCM plates and advanced insulation designs, has significantly improved the performance of refrigerated containers. These PCM-based plates, which can be applied to the interior, exterior, or both sides of the container walls, enhance passive cooling efficiency. M.A. Ben Taher et al. [6] provided a comprehensive review of computational and experimental advancements in refrigerated truck technologies, emphasizing the role of insulation and cold storage systems in optimizing energy efficiency and temperature control. Similarly, T.B. Umate et al. explored both active and passive PCM approaches, studying the impact of PCM with varying melting points in Cold Thermal Energy Storage (CTES) units across different applications. These advancements highlight the synergy between simulation, experimental studies, and PCM integration in cold chain logistics. Similarly, Michele Calati et al. [7] presented the overview on use of low temperature PCM in the refrigerated transport application. The low temperature PCM is used in the internal and external walls of refrigerated containers. Jouhara et al. [8] reviewed various low-temperature energy storage (LTES) technologies and their applications, considering methods of thermal energy storage (TES), which include sensible, latent, and thermochemical storage. However, this study did not specifically explore the use of phase change materials (PCM) in cold chain applications. Nie et al. [9] carried out simulation of cold thermal energy storage applications to study the effect of materials and design methodologies used in latent heat storage system. But only a limited number of applications were discussed on cold chain transportation. Zhao et al. [10] discuss advancements in cold storage technology for cold chain transportation and distribution, highlighting several proposed designs and analyzing temperature distribution characteristics within refrigerated vehicles. The study by Michele Calati et al. [11] focuses on the CFD analysis and development of an innovative latent thermal energy storage (LTES)

system tailored for food transportation. The research highlights the role of phase change materials (PCM) in enhancing thermal regulation during transit. Simulation results reveal that the proposed LTES system effectively maintains stable temperatures within the storage unit, crucial for preserving food quality. The CFD analysis provides insights into heat transfer mechanisms and PCM behavior, ensuring optimal design and energy efficiency. The study underscores the potential of PCM-based LTES in reducing energy consumption and improving the reliability of food transportation under varying operational conditions. A study conducted by Yin et al. [12] on an incubator with XPS foam and PCM layers found that the PCM layer improve holding time by 99 times compared to XPS alone, and a combined XPS-PCM system provide 14.7 % longer protection. The optimal configuration for vaccine transportation is 7.5 cm thickness for PU/XPS/PCM, which ensure 72 h of thermal protection.

Luca Cirillo et al. [13] analyze the effect of E-26 and E-29 PCMs in refrigerated trucks on temperature control, phase change fraction, costs, and energy savings. Using a transient model, the effectiveness of PCM-based systems in maintaining -18°C under various conditions was evaluated. The results show that PCM incorporation significantly enhances temperature stability, maintain uniform temperature, extends the shelf life of perishable products and reduces energy costs up to 79 %. Similarly, Song et al. [14] developed a simulation model for refrigeration systems using TRNSYS and MATLAB, and optimize the energy consumption of the chiller's key operational parameters using genetic algorithms. The results indicate a significant reduction in total energy consumption, with an average savings of 0.5 kW, and an increase in the Coefficient of Performance (COP) by an average of 0.15, demonstrating improved efficiency of the refrigeration system. Liu et al. [15] study proposes a novel phase-change cold storage unit (PCCSU) for mobile refrigeration in transport vehicles to enhance temperature control. The PCCSU installed at the front of the insulated compartment is charged during off-peak periods and used to cool the compartment. The results indicate that the PCCSU significantly improves temperature control, supporting multiple temperature settings, unlike conventional single-temperature systems. In hot summer conditions, the PCCSU maintains compartment temperatures at -12.3°C , -14.5°C , and -16.5°C for 16.6 $^{\circ}\text{C}$, 14.7 $^{\circ}\text{C}$, and 10 h, respectively, with an average ambient temperature of 29°C . It lowers the minimum internal temperature by 3°C and extends control time by 7.4–9.4 h compared to traditional systems. Additionally, the PCCSU enhances heat transfer efficiency and reduces energy costs by 15.4 %–91.4 % compared to diesel-driven refrigeration units, significantly improving the safety and efficiency of refrigerated transport. The study by Mihir Madhu et al. [16] investigates the use of N-tetradecane phase change materials (PCM) to enhance cold chain logistics for refrigerated trucks. The findings indicate that N-tetradecane PCM effectively reduces energy consumption by maintaining stable temperatures within the truck compartment, achieving up to a 25 % reduction in refrigeration system energy usage. Moreover, PCM helps smoothen temperature fluctuations during power outages, ensuring consistent cold storage conditions. The inclusion of PCM is shown to extend the temperature hold duration by over 40 % and acts as a buffer, maintaining cold temperatures for up to 6 h during power interruptions. Experimental data validated the numerical simulations, demonstrating the reliability of N-tetradecane PCM as a robust thermal storage medium.

The walls of refrigerated containers, due to their extensive surface area, often serve as a conduit for heat infiltration into the cold chamber. Adekomaya et al. [17] proposed an innovative approach to reduce heat infiltration into refrigerated containers by replacing the external metallic sheet with a lightweight and low thermal conductivity material. Few studies have investigated incorporating a PCM within the standard PU insulation walls of refrigerated vehicles. Same studies have shown that incorporating a PCM layer within the existing polyurethane (PU) insulation of refrigerated vehicle walls absorb and release heat as it changes phase, and regulate the internal temperature [18–20]. Tinti

et al. and Michel et al. [21,22] created a composite layer where PCM is micro-encapsulated within the PU insulation. This method integrates PCMs directly into the insulation material, enhancing thermal management without the need for a separate layer. The study conducted by Ali Mousazade et al. [23] evaluates the thermal performance of cold panels embedded with phase change materials (PCM) in refrigerated trucks. The research emphasizes the effectiveness of PCM in maintaining consistent temperatures during transportation. Results indicate that the integration of PCM significantly enhances thermal regulation, extending temperature stability during periods without active refrigeration. The study demonstrates that cold panels with PCM can reduce energy consumption and improve the overall efficiency of refrigerated transport systems, making them a sustainable solution for cold chain logistics.

Tinti et al. [21] explored the incorporation of micro-encapsulated phase change materials (PCMs) into polyurethane (PU) foam using a dispersion process. This study focused on designing a composite layer that integrates PCM into PU insulation panels. The thermographic analysis was carried out to assess the effect of varying PCM weight percentages on thermal performance under directional heat flow. In a related study, Michel et al. [22] investigated a similar composite layer design but applied it differently. The location and thickness of the PCM-PU composite layer was varied relative to the standard insulation layer in refrigerated vehicles. The experiments and numerical modeling revealed that placing the PCM-PU composite layer closer to the external wall result in a 4.1 % reduction in heat transfer. This suggests that strategic placement of the PCM-PU layer can enhance thermal insulation performance in refrigerated transport systems. Glouannec et al. [18,24] studied superinsulation materials, incorporating aerogel and reflective metal foil for the first time with polyurethane foam, polyester, and glass fibre within refrigerated vehicle panels. This is referred to as Case 1, a multilayer panel. Case 2 consists of a 5 mm Energain PCM layer composed of 60 % paraffin wax. The thermal behavior of these multilayer configurations was compared to a reference case. The average decrease in energy consumption for Case 1 and Case 2 was 29 % and 25 % respectively, compared with the reference case. Further experiments and simulations were conducted to investigate the effect of solar irradiation on the outer wall side of the reference case and Case 1. According to the test results, the aerogel and reflective metal foil layers limit the increase in heat flux density due to solar irradiation. Fioretti et al. [19] used a Rubitherm RT35HC paraffin wax layer on the external wall of a cold storage room in Ancora for two days under real environmental conditions. The average heat transfer reduction of 5.55 % and 8.57 % with peak delays of 4.30 h and 3.30 h respectively was observed. During the day, incoming solar irradiation is absorbed by the external surface, transferring heat inside the cold chamber. The PCM layer absorbs the incoming heat flux during its melting phase, acting as a thermal energy storage medium. Under extreme weather conditions in three different towns in Italy, Copertaro et al. [25] investigated the effect of several types of PCMs selected according to climate conditions, despite their different melting temperatures ranging from 27.5 °C to 46.5 °C and varying latent heats. According to ISO standards, a 20-ft refrigerated container was considered in the numerical simulation using the COSMOL Multiphysics software. Two configurations were studied: the first was a reference case container with 100 mm polyurethane foam sandwiched between steel sheets, and the second included an additional 30 mm PCM layer in a standard insulated container. It was reported that, under Italian summer conditions, RT35HC paraffin wax showed the most promising results among the PCMs tested.

M. Calati et al. [20] carried out simulations to assess the thermal performance of a composite wall under typical summer conditions in Italy. The composite wall design includes a traditional polyurethane (PU) insulation layer and an additional Phase Change Material (PCM) layer thickness (0.5 cm to 2 cm thick) inside the refrigerated container. PCMs absorb or release heat during phase change. Three paraffin-based PCMs (RT2HC, RT4, and RT5HC) were used. The results showed that a 0.5-cm PCM layer maintain air temperatures below the PCM solidus

temperature (the temperature below which PCM remain solid) for four hours. Afterward, temperatures raise, risking high-sensitivity foodstuffs. The 1 cm PCM layer maintains air temperatures 2 °C above the melting point for 10 h. The 2 cm PCM layer provided the most consistent temperature control, with only 25 % of the PCM melting, indicating a potential for longer transport durations. Similarly, the novel design of refrigerated walls was carried out by Ahmed et al. [13] and an experimental study on the standard refrigerated walls integrated with phase change materials was carried out. A thin copper tube filled with paraffin-based PCM (melting temperature 7 °C) was used. The experiment result shows that the average heat transfer rate range between 11.3 % and 43.8 %. Moreover, there is an average reduction in peak heat transfer of 29.1 % and an average energy savings of 16.3 % compared to a standard refrigerated wall.

Wu et al. [4,26] introduced an innovative carbon footprint model to evaluate the environmental impact of refrigerated trucks. This model comprehensively considers the life cycle impacts of the refrigeration system, the insulated truck body, and the truck chassis. Their study underscores the significant carbon footprint associated with refrigerated trucks, particularly during the usage phase. Remarkably, this phase accounts for approximately 96 % of the total greenhouse gas (GHG) emissions attributed to refrigerated trucks. When refrigerated vehicle containers are exposed to solar radiation during the summer, internal temperatures can significantly increase. Additionally, frequent opening and closing of the door during transportation further increases internal temperatures. The air recirculation and the integration of PCM significantly impact exergy components, optimizing the system's performance. Study conducted by Hadi samimi et al. [27,28] employing PCM in conjunction with air recirculation reduce exergy destruction, leading to enhance drying efficiency. Similarly, it is emphasized that these strategies not only improve energy utilization but also contribute to more sustainable drying operations by minimizing thermal losses [29,30].

Taher et al. [31] carried out simulations using Unsteady Reynolds-Averaged Navier-Stokes (URANS) to investigate the aerodynamic and thermal performance of refrigerated trucks under various conditions. This study discusses the effect of air curtains on energy consumption and cooling efficiency when truck doors are opened under different loading conditions. The results show that air curtains are highly effective in optimizing both energy consumption and the duration needed for cooling. Specifically, the use of air curtains leads to energy savings up to 25.3 % in energy consumption. This significant reduction highlights the practical benefits of integrating air curtains into the refrigerated trucks to enhance energy efficiency and reduce operational costs.

Temperature fluctuations during loading and unloading are major issues in food transportation, affecting quality and the preservation of perishable goods. To address this issue, researchers globally have been investigating passive low-temperature distribution methods, i.e., passive thermal energy storage systems with low melting temperature phase change materials. One of the researchers Taher et al. [32] conducted a study on the performance of refrigerated trucks under two approaches: with and without the inclusion of Phase Change Materials (PCMs). The PCMs are integrated into a module containing several baffle-type panels placed in the truck roof, a novel method for storing and later releasing cold energy in refrigerated trucks. The study tested three paraffin-based PCMs—RT3 HC1, RT5, and RT8 HC—to determine the most effective material for maintaining the desired temperature range. The results demonstrate significant reductions in cooling energy required when the truck door is opened for loading or unloading: 61 % for RT3 HC1, 57 % for RT5, and 35 % for RT8 HC. Additionally, air temperature fluctuations inside the refrigerated trucks after turning off the refrigerated system are reduced, maintaining overall uniform temperature. Similar, Zhangyang Kang et al. [33] investigates the integration of phase change materials (PCMs) into refrigerated truck boxes to enhance cooling efficiency during cold chain transportation. The results reveal that incorporating PCMs reduced temperature fluctuations within the truck by approximately 4 °C, with RT3HC PCM demonstrating superior

performance in maintaining air temperature stability. Increasing the number of PCMs improved heat transfer efficiency and solidification/melting rates. Notably, a configuration with nine PCMs achieved a 57.13 % higher heat transfer efficiency compared to a setup with three PCMs. Implementing a flow-guiding PCM arrangement enhanced temperature uniformity and improved solidification efficiency by 3.8 % over elongated strip-shaped PCM configurations.

Despite significant advancements in refrigerated container technology, maintaining a consistent temperature distribution while minimizing energy consumption remains a challenging task. Phase Change Materials (PCM) are widely recognized for their latent heat storage capabilities; however, detailed investigations into their thermal behavior in refrigerated containers under realistic operating conditions characterized by fluctuating temperatures and dynamic cooling loads are still limited. Previous studies have largely focused on improving thermal insulation or utilizing PCM for temperature regulation in isolation. These approaches often neglect the combined effects of PCM placement and airflow dynamics within the container. Furthermore, limited attention has been given to integrating PCM plates at strategic locations to enhance both thermal storage and airflow distribution. The potential of PCM plates to act as flow guides, improving cold air circulation and ensuring uniform temperature distribution, has not been fully explored. This underscores the need for innovative designs that can simultaneously optimize temperature stability and energy efficiency.

To address these gaps, this study investigates the inclusion of PCM plates at critical locations on the roof and the left and right-side walls of a refrigerated container. This approach aims to mitigate temperature fluctuations and enhance temperature stability, especially for high-sensitivity food products. Beyond thermal storage, the PCM plates in this study are designed to act as flow guides, improving the distribution of cold air throughout the container. By analyzing the combined effects of PCM integration and air circulation dynamics, this numerical investigation offers a comprehensive solution to overcome the limitations of existing systems, providing a significant advancement in the field of refrigerated container technology.

2. Numerical method

2.1. Geometrical model of refrigerated container

Medium-duty refrigerated vehicles are used for ground transportation of perishable foods. The container construction consists of standard insulated (Polyurethane foam) sandwich panels reinforced with steel to ensure durability and strength. These insulation panels, which typically range in thickness from 50 to 150 mm, are specifically designed for short-haul refrigerated transportation, compared to long-haul trucks or tractor-trailers. In this study, the insulated sandwich panels used for the medium-sized refrigerated trucks have a thickness of 51.7 mm, with polyurethane foam insulation acting as the core material. The panels consist of galvanized outer layers and inner panels are made of food-grade stainless steel. For aesthetic appeal, the outer layer of the refrigerated truck is coated with white automotive paint ES-170. Table 1 presents the composition of the wall surfaces of the refrigerated truck. The dimensions of the refrigerated truck container in this study are 3 m in length, 1.6 m in width, and 1.7 m in height. The refrigeration system

dimensions are 1 m in length, 0.4 m in width, and 0.2 m in height. The THERMAL MASTER T-1400 refrigeration equipment is used in the study, which has a cooling capacity of 4689 W at 30 °C and 2400 RPM. The system under investigation only experiences cooling, implying that the refrigeration equipment cannot reach the phase transition temperature and that no phase transition occurs within the system. Fig. 1 shows a simplified model of a medium-sized refrigerated vehicle container [34]. The diagram shows four temperature monitoring points: point 1 at the air inlet, point 2 at the outlet, point 3 at the door, and point 4 on the cabinet floor. The location coordinates of each point are shown in Table 2.

2.2. Model related assumptions

This study investigates the thermal behavior of a medium-sized refrigerated container with PCM-based thermal energy storage. To streamline the modeling process and focus on the primary objective, several simplifying assumptions are employed. First, the intricate airflow within the refrigeration system is excluded. While airflow dynamics contribute to real-world heat transfer, the study prioritizes conduction and convection mechanisms, which are the dominant modes of heat transfer under steady-state conditions. Radiative heat transfer is also neglected, as the reflective interior surfaces and low operating temperatures reduce its influence. This simplification is commonly adopted in similar thermal studies. Additionally, external airflow over the container's walls during motion is assumed to be uniform, and air infiltration through gaps is not considered. Although these factors could introduce localized variations, their overall impact on PCM performance is expected to be minimal.

The model further adopts an open system approach, utilizing the enthalpy difference between the air inlet and outlet to approximate a closed system, a method validated in prior studies for its balance of accuracy and computational efficiency. Air inside the container is treated as a Newtonian, incompressible fluid with constant thermophysical properties. This assumption aligns with standard practices, given that temperature variations within the operational range have minimal impact on fluid properties.

2.3. Mathematical model

The present study analyses two modes of operation for the

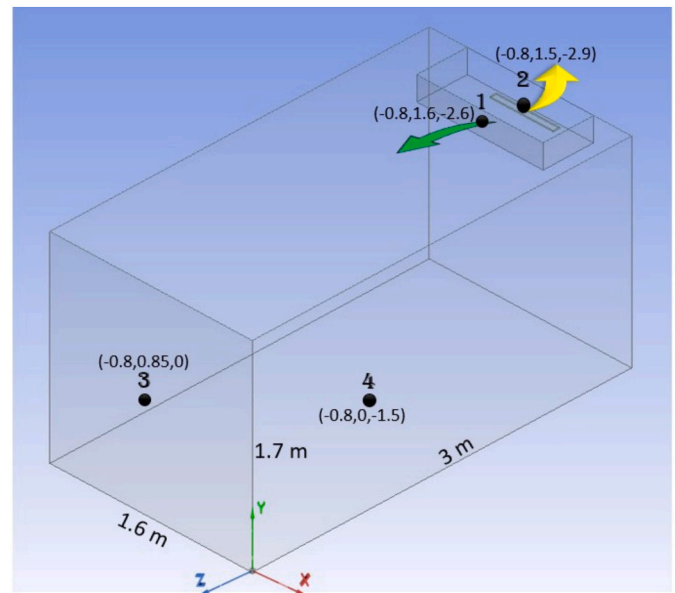


Fig. 1. Schematic of refrigerated container with analysis points.

Table 1
Material composition of container walls of refrigerated vehicle. [34].

Types of walls	Construction materials	Thickness (mm)
Floor wall	– Stainless steel sheet (AISI 304)	0.7
	– Galvanized steel sheet	1
	– Standard polyurethane foam sheet	50
Roof and side wall	– Galvanized steel sheet	1
	– Stainless steel sheet (AISI 304)	0.7
	– Standard polyurethane foam sheet	50
	– Automotive (ES 170) paint	0.1

Table 2

Coordinate of the analysis points.

Points	X (m)	Y (m)	Z (m)
1. Air inlet	-0.8	1.6	-2.6
2. Outlet	-0.8	1.5	-2.9
3. Door	-0.8	0.85	0
4. Cabinet floor	-0.8	0	-1.5

refrigerated vehicle. The first mode involves a forced turbulent refrigeration process driven by the refrigeration unit. The second mode involves natural convection occurring when the vehicle doors are open.

Turbulence equations utilize Eddy's viscosity models: κ - ϵ and κ - ω . The κ - ϵ model, designed for high Reynolds numbers, analyses the flow within the turbulent core. In contrast, the κ - ω model, which uses two equations, is tailored for near-wall flow at low Reynolds numbers. Menter's [35,36] Shear Stress Transport (SST) κ - ω model, an enhancement of the standard κ - ω model, accurately predicts indoor air conditions. Inside the refrigerated space, air is treated as a compressible fluid with variable properties. However, simulation results show that variations in air density have a negligible impact on accuracy in this context. Heat transfer within the container is driven by conduction and convection, while external radiation effects are disregarded due to their insignificant role in forced convection.

The FLUENT solver is used to derive the SST model equations. The continuity equation, a fundamental principle stating that mass is conserved and not created or destroyed within a fluid, is always met. This principle is mathematically represented in differential form.

For the continuity Eq. (1) in a refrigerated system in a refrigerated vehicle,

$$\frac{\partial \rho_f}{\partial t} + \nabla \cdot (\rho_f u_i) = 0 \quad (1)$$

For a refrigerated system in a refrigerated vehicle, the momentum Eqs. (2)–(4) needs to account for the forced turbulent refrigeration process and natural convection.

$$\frac{\partial (\rho_f u_i)}{\partial t} + \nabla \cdot (u_i u_j) = -\nabla p + \nabla \cdot (\mu (\nabla u_i + (\nabla u_j))) + \rho_f \cdot g + S_{refrigeration} + S_{natural} \quad (2)$$

Considering the forced and natural convection:

1. Forced Convection (Refrigeration Unit On):

$$\frac{\partial (\rho_f \cdot E)}{\partial t} + \nabla \cdot (u_i (\rho_f E + p)) = \nabla \cdot (k_{eff} \nabla T) + S_{refrigeration} \quad (3)$$

2. Natural Convection (Doors Open):

$$\frac{\partial (\rho_f \cdot E)}{\partial t} + \nabla \cdot (u_i (\rho_f E + p)) = \nabla \cdot (k_{eff} \nabla T) + S_{natural} \quad (4)$$

For a fully air domain,

$$k_{eff} = k_{air}$$

where:

- $S_{refrigeration}$ is the source term representing the effect of the refrigeration unit.
- $S_{natural}$ is the source term representing the effect of natural convection when the doors are open.

2.3.1. k - ω SST model

The k - ω Shear Stress Transport (SST) model Eqs. (5) and (6) are popular turbulence models used for its robustness in predicting flow

separation and near-wall flows, making it suitable for analyzing a refrigerated vehicle where both forced and natural convection can occur.

$$\frac{\partial (\rho k)}{\partial t} + \nabla \cdot (\rho k u) = p_k + \beta^* \rho k \omega + \nabla \cdot ((\mu + \sigma_k \mu_t) \nabla k) \quad (5)$$

$$\begin{aligned} \frac{\partial (\rho \omega)}{\partial t} + \nabla \cdot (\rho \omega u) = & \nabla \cdot ((\mu + \sigma_\omega \mu_t) \nabla \omega) + \frac{\gamma}{\mu_t} P_{SST} + \beta \rho \omega^2 \\ & + 2(1 - F_1) \frac{\sigma_\omega \rho_f}{\omega} \nabla k \cdot \nabla \omega \end{aligned} \quad (6)$$

2.3.2. For refrigerated container walls

Eq. (7) describes the heat conduction and heat storage within the wall [32,37].

$$\rho_w C_{pw} \frac{\partial T_w}{\partial t} + \nabla \cdot (\lambda_w \nabla T_w) = 0 \quad (7)$$

where λ_w , ρ_w , C_p , w , T_w are thermal conductivity, density, specific heat, and temperature of the wall.

2.3.3. For phase change material

The enthalpy method is a popular approach to describing heat transfer in phase-change materials, as described in Eqs. (8)–(10) [38,39].

$$\rho_{pcm} C_{p,pcm} \frac{\partial T_{pcm}}{\partial t} + \rho_{pcm} \frac{\partial H_{pcm}}{\partial t} = \nabla \cdot (k_{pcm} \cdot \nabla T_{pcm}) \quad (8)$$

The enthalpy H_{pcm} of the PCM is defined as:

$$H_{pcm} = \rho_{pcm} [C_{p,pcm} (T_{pcm} - T_{ref}) + L \phi(T_{pcm})] \quad (9)$$

The phase fraction ϕ can be defined as:

$$\phi(T_{pcm}) = \begin{cases} 1 & \text{if } T_{pcm} < T_{solidus} \text{ (Solid phase)} \\ 0 & \text{if } T_{pcm} > T_{liquidus} \text{ (Liquid phase)} \\ \frac{T_{pcm} - T_{solidus}}{T_{liquidus} - T_{solidus}} & \text{if } T_{solidus} \leq T_{pcm} \leq T_{liquidus} \text{ (Phase change region)} \end{cases} \quad (10)$$

2.3.4. Energy efficiency (η)

The energy efficiency (η) of a Phase Change Material (PCM) system, is given by Eqs. (11)–(12),

$$Q_r = \int_{V_f} \rho_f C_{p,f} (T_f - T_0) dv_f \quad (11)$$

$$\eta = 1 - \frac{Q_r}{Q_{r0}} \quad (12)$$

where, Q_r = Recovery energy with PCM, Q_{r0} = Recovery energy without PCM (i.e., through the open door).

2.4. Numerical simulation solver setup

2.4.1. Initial conditions

The 3D geometry of the refrigerated vehicle is created in the Design Modeler in the ANSYS workbench 2021 R1. The computer system used in this simulation was an Intel (R) Xeon (R) W-2275 CPU at 3.31 GHz with an installed RAM of 64.0 GB and a 64-bit operating system with an x64-based processor. The computer system has 14 cores, with 12 of them dedicated to running the simulation using precision parallel processing. Fig. 2 shows the cold air inlet (1) and air outlet (2) with points on the door (3), and floor (4). The specified domain allows for establishing conditions and configurations related to fluid properties and boundary conditions to comprehend fluid behavior within the system. The software accepts direct inputs for air properties such as density and viscosity.

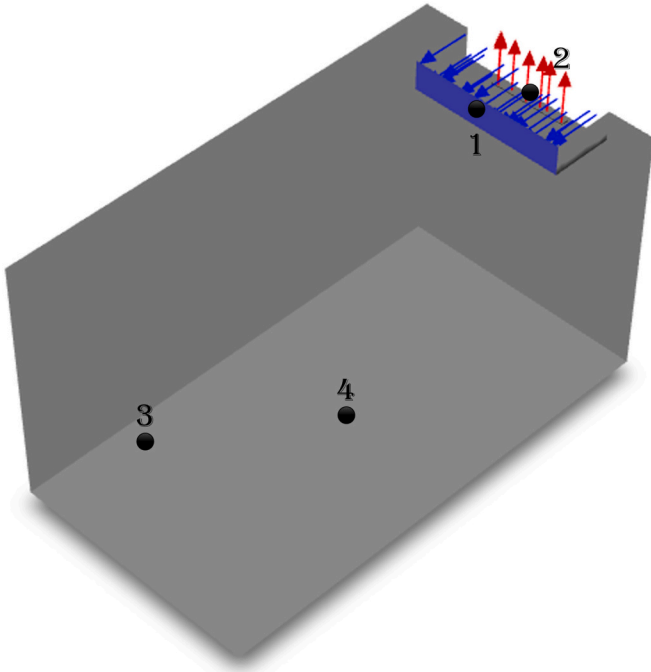


Fig. 2. 3D Model of refrigerated vehicle container.

2.4.2. Boundary conditions

2.4.2.1. At entrance. Considering that the temperature of the refrigerated vehicle is not constant, the refrigeration equipment inlet temperature must be input into the simulation software as a temperature versus time function, necessitating dynamic simulation using Fluent. In this study, the inlet temperature is dynamically modeled using experimental data fitted with an exponential function (Eq. (13)) [26].

To visualize the temperature profile, temperature is defined by Eq. (13):

$$T(t) = 273.15 + 11.95 \cdot e^{(-0.001058 \cdot t)} + 11.58 \cdot e^{(-0.01036 \cdot t)} \quad (13)$$

A User-Defined Function (UDF) for ANSYS Fluent that controls the refrigerated air temperature is based on a given profile Eq. (13). This UDF integrates custom temperature control logic into the simulation, enabling the UDF to regulate the inlet temperature as specified.

2.4.2.2. At walls. The study utilizes air, aluminum, and polyurethane foam panels. In the computational domain settings, the fluid domain is defined as air, while the solid domain represents the insulation layer. The boundary conditions account for convective heat transfer between the external wall surface of the refrigerated truck and the external air, excluding external radiative heat transfer. For the non-slip wall surface, a convective boundary condition with external surface temperature is shown in Table 3, with the convection coefficient calculated for each wall surface as $3.41 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ [34]. The wall material is designated as polyurethane foam board. Since the cold dissipation of the wall surface of the refrigeration equipment is not considered, a heat flux boundary

Table 3
Wall temperature [34].

Different wall	External temperature (K)
Left side wall	296.3
Right side wall	294.6
Roof	304.5
Floor	296
Front wall	303
Door	295.2

condition is set for this wall surface, with the heat flux set to 0 W/m^2 , effectively making it an adiabatic wall surface.

3. Numerical model validation

3.1. Mesh study and convergence verification

In computational fluid dynamics (CFD) simulations, ensuring mesh refinement is crucial for obtaining accurate and reliable results. Fig. 3 presents the unstructured mesh model, consisting of 550,953 elements and 97,605 degrees of freedom (DOF). The mesh study, often referred to as a grid convergence study, systematically refines the mesh to evaluate the solution's sensitivity to mesh size. Fig. 4 demonstrates the results of this mesh refinement process by monitoring the entrance temperature across mesh sizes of 135,868, 319,949, 550,953, and 1,078,372 elements. The objective of this analysis is to determine the mesh size at which further refinement has a negligible effect on the results. As shown in Fig. 4, the temperature profiles for the 550,953-element mesh closely match those of the 1,078,372-element mesh, with minimal deviations. This convergence verifies that the solution is effectively independent of the mesh size beyond 550,953 elements. Thus, this mesh size provides an optimal balance between computational efficiency and result accuracy for the given simulation.

3.2. Comparison with experimental data

3.2.1. Model validation

Fig. 5 compares the aerothermal behavior of different URANS unsteady turbulence models (standard $\kappa - \omega$, SST $\kappa - \omega$, standard $\kappa - \epsilon$) with experimental data for the air temperatures inside a refrigerated truck. The validation process shows that all three models closely match the experimental measurements, indicating their reliability in predicting aerothermal behavior. Among the models, the SST $\kappa - \omega$ model demonstrates superior accuracy due to its ability to combine the strengths of both $\kappa - \omega$ and $\kappa - \epsilon$ models. This combination allows it to accurately predict airflow and temperature variations both near and far from the refrigerated container walls. The root mean square error (RMSE) values confirm this, with the SST $\kappa - \omega$ model showing the best agreement with experimental data, as shown in Table 4. To validate the internal

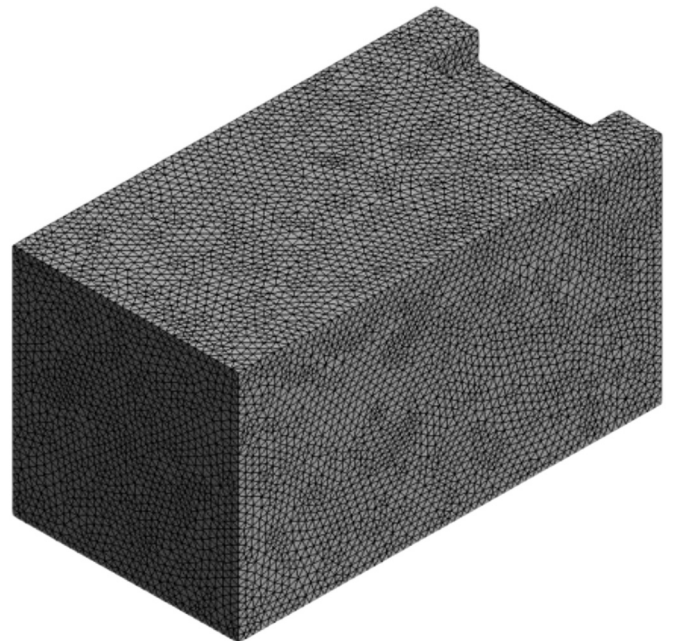


Fig. 3. Computational mesh for refrigerated container simulation.

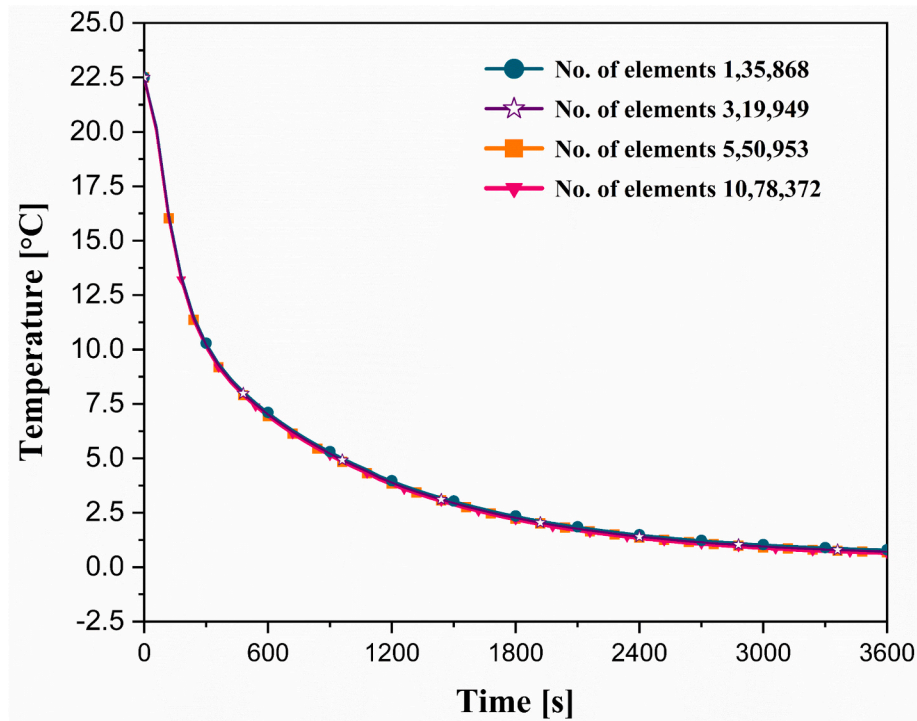


Fig. 4. Mesh sensitivity analysis.

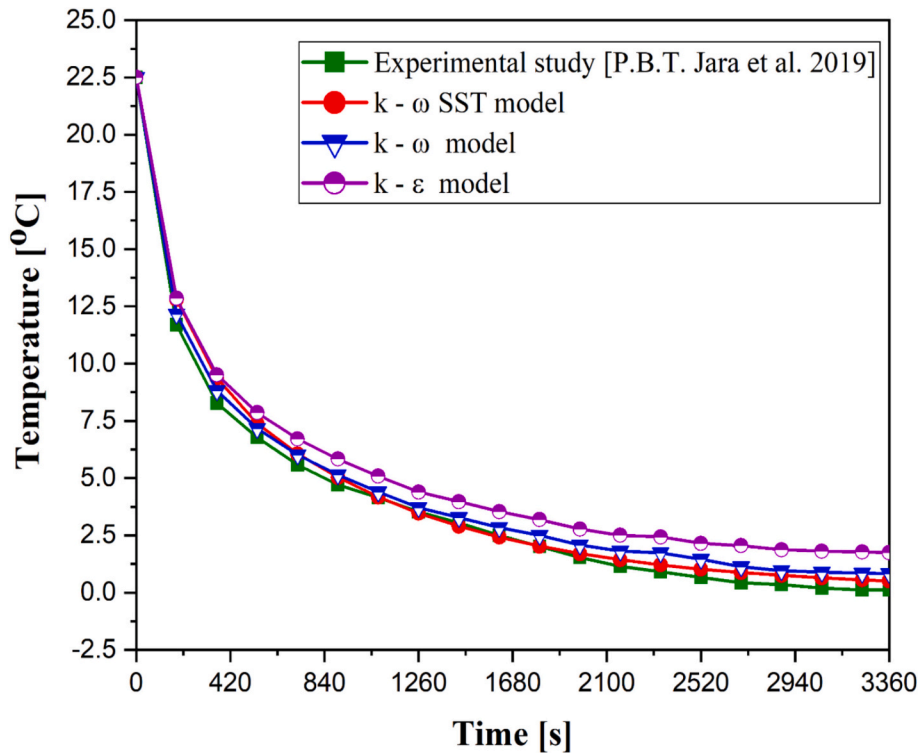


Fig. 5. Entrance temperature validation using different turbulence models.

temperature analysis model of the refrigerated vehicle, comparisons were made between the experimental study as well as the numerical study conducted by P.B.T. Jara et al. [34] and the present computational results. This validation is performed through graphical analysis and absolute error calculations, which describe the maximum temperature deviation between the two sets of results. The maximum absolute errors

observed are -1.53°C at the entrance, -1.13°C at the exit, 0.42°C on the floor, and 0.69°C at the gate, as detailed in Table 4.

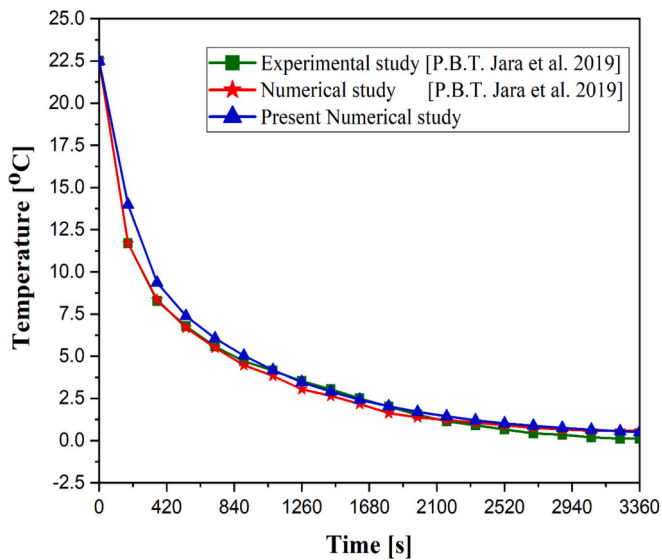
The data lines in Figs. 6 to 9 show the temperature behavior over time. The present study data line shows the temperature variation curve obtained from the simulation, along with the experimentally measured values and the numerical simulation results from P.B.T. Jara et al. [34].

Table 4

Absolute temperature deviations at various points in refrigerated container.

Time (s)	Entrance	Exit	Floor	Gate
0	0.00	0.00	0.00	0.00
180	-1.53	0.23	0.26	0.46
360	-0.83	-0.16	0.32	0.42
540	-0.69	-0.14	0.27	0.49
720	-0.63	-0.59	0.15	0.47
900	-0.66	-0.78	0.42	0.32
1080	-0.47	-0.94	-0.10	0.42
1260	-0.57	-1.05	-0.03	0.52
1440	-0.42	-0.87	-0.08	0.65
1620	-0.43	-0.99	-0.03	0.40
1800	-0.60	-1.13	0.13	0.42
1980	-0.52	-1.04	-0.03	0.45
2160	-0.42	-0.97	-0.01	0.68
2340	-0.37	-0.92	0.13	0.36
2520	-0.35	-0.84	0.18	0.50
2700	-0.35	-0.64	0.18	0.43
2880	-0.32	-0.30	0.18	0.69
3060	-0.29	-0.24	0.15	0.50
3240	-0.21	-0.18	0.18	0.48
3360	-0.16	0.11	0.09	0.69
Average	-0.49	-0.57	0.12	0.47

The bold-highlighted values in the table represent the Root Mean Square (RMS) deviation in temperature at the entrance, exit, gate, and floor. The RMS deviation quantifies temperature fluctuations at these key locations. These values are significant for assessing thermal stability within the refrigerated container. A higher RMS deviation indicates greater fluctuations, suggesting localized heat transfer effects.

**Fig. 6.** Temperature data line comparison at the entrance.

Figs. 6 to 9 shows, the temperature behaviors at each considered inner point of the refrigerated vehicle container. Fig. 6 shows the temperature data line at the entrance, representing the air inlet temperature to the refrigerated vehicle. The temperature varies as a function of time and is represented by an exponential function, resulting in an average temperature variation of -0.49°C . At Fig. 7 shows the air outlet temperature of the system. The temperature variation follows a similar trend of the experimental and numerical curves, with an average temperature difference of -0.57°C . Fig. 8 shows the temperature variation of the vehicle floor, where the lowest average temperature variation of 0.12°C was observed. Finally, Fig. 9 shows the temperature variations at the door, with an average temperature variation of 0.47°C .

4. Numerical analysis and results

The model is validated by comparing experimental and simulation data, showing its consistent predictive accuracy. This allows us to investigate how temperature distribution is affected by the frequent opening and closing of doors during the loading and unloading of refrigerated vehicles. Initially, the study focuses on the air temperature inside the refrigerated container without PCM, and the cooling system ON for 120 min, followed by the system being OFF during unloading. Subsequently, PCM-filled plates are installed in the refrigerated container along the left and right side walls. Additionally, the study examines the flow-guiding PCM plates installed on the roof of the refrigerated container. The main objective is to compare the temperature reduction and distribution inside the container with a conventional refrigerated container.

4.1. Thermal analysis of door opening and closing processes in a refrigerated vehicle

Once the model is validated, the thermal performance of the closed and open-door refrigerated vehicles is studied through a proposed scenario. The computational fluid dynamic SST k- ω model was implemented after confirming its accuracy in representing the aerothermal behavior of refrigerated vehicles. The scenario involves a single cycle, where the cooling system operates for 120 min with the door closed, and then for 30 min with the door open for loading and unloading Fig. 10.

The aerothermal dynamics of the refrigerated container during both closed and open-door phases was studied. Fig. 11 shows the temperature variation throughout one cycle at various times, excluding the presence of an installed PCM plate. Initially, the temperature inside the container is around 22.5°C (Fig. 12a). During the cooling period (with the door closed), the coldest temperatures are observed near the roof, while the temperature decreases in other areas of the refrigerated container. As the door remains closed for the first 120 min, the temperature rapidly drops. This decline is sharp and the temperature reach around 0.67°C after approximately 3600 s (1 h). The cooling continues more gradually, and by the end of 120 min (7200 s), the temperature stabilizes just near 0°C (Fig. 12b). During this period the container maintain a low temperature effectively with the door closed. While the door remains closed (120 min), the temperature remains relatively stable, oscillating slightly around 0°C , ensuring the preservation of the frozen foods. After 7200 s, the door is opened for 30 min (1800 s), causing a rapid increase in temperature due to the ingress of warmer external air. The temperature rises sharply from 0°C to around 10.69°C within one minute of the door being opened. Then it reaches 22.2°C by the time the door is closed (Fig. 12c).

Fig. 12a shows the initial temperature distribution within the refrigerated container during the cooling phase. The temperature varies between 273.15 K (0°C) near the cooling unit and 297.88 K (24.73°C) in the warmer regions. The refrigerated container has limited thermal regulation, where heat accumulates in the upper and corner areas due to natural convection and stratification. Fig. 12b ($t = 3600$ s) shows progress in cooling process. The plot shows that air temperature is still exhibiting hotspots in areas far from the cooling source, particularly near the center and far corners. The thermal capacity depends on the insulation properties of the polyurethane, resulting in inefficient cooled air distribution. Fig. 12c ($t = 7260$ and $t = 9000$ s) shows the effect of door openings. At $t = 7260$ s, warm air infiltrates quickly through the door opening (left side), forming a clear hot air plume. This dynamic flow behavior causes a sudden increase in temperature in the front zone, with heat increasing and moving toward the container's rear. At 9000 s, the contour plot shows the hot air plume inside the refrigerated container. Warm air enters through the door and accumulates in the container's upper sections, particularly near the ceiling and rear corner. In Fig. 12c, the probe point marked inside the blue circle is strategically

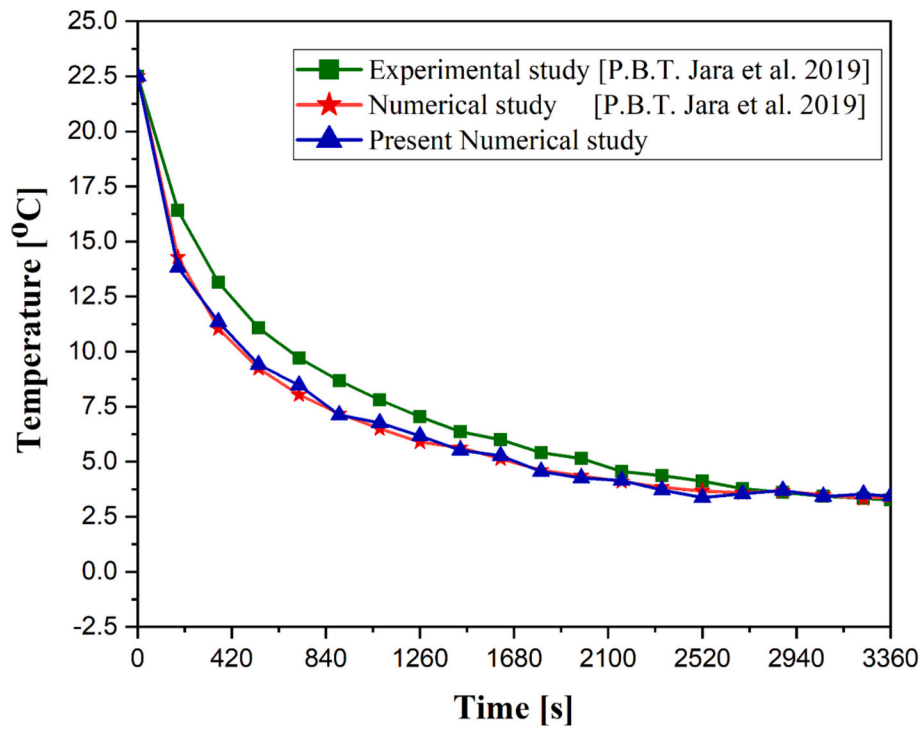


Fig. 7. Temperature data line comparison at the outlet.

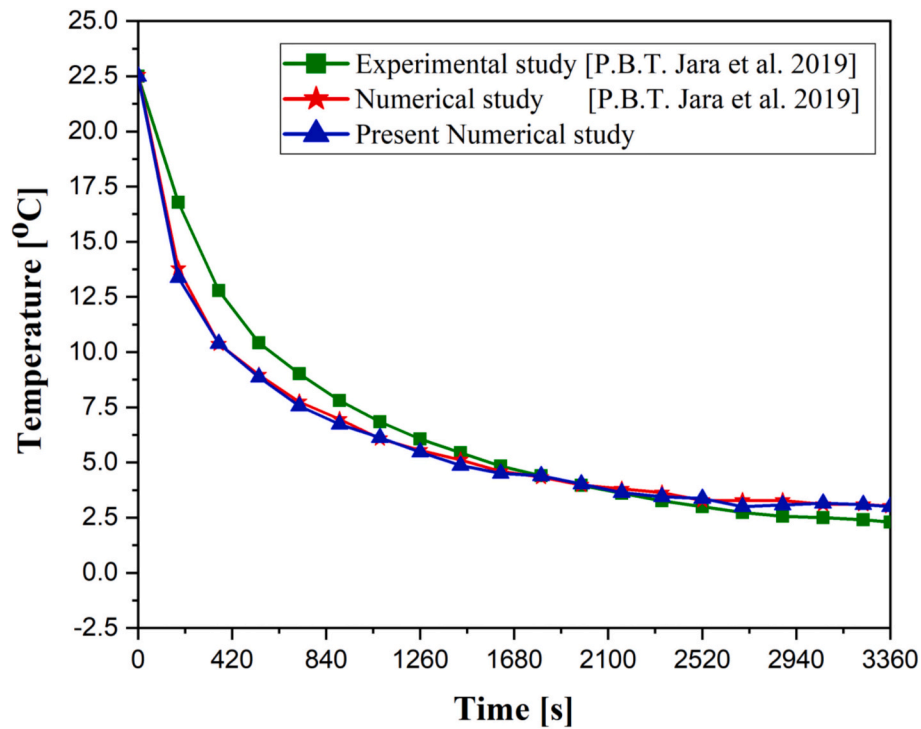


Fig. 8. Temperature data line comparison at the floor.

placed to measure both the air velocity and temperature ingress into the refrigerated container, specifically to observe the effects of the door opening. The data reveals that warmer air from the external atmosphere enters the container when the door is opened. The temperature of this incoming air is recorded as 24.58 °C, and it enters at a very slow velocity of 0.013 m/s. The slow infiltration rate indicates that while the door is opened the warmer air gradually enters the container, which can cause

an increase in the internal temperature. The cooling system immediately counteract the increase in internal temperature within the container as the polyurethane-insulated container relies on the cooling unit for thermal regulation. To mitigate these temperature fluctuations, this study incorporates PCM-filled plates installed on the side walls and ceiling of the container.

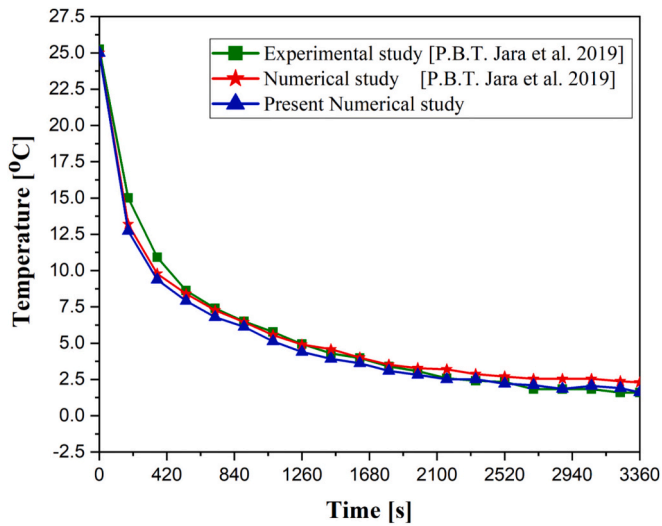


Fig. 9. Temperature data line comparison at the door.

4.2. Influence of PCM incorporation on temperature distribution in refrigerated containers

4.2.1. Incorporation of PCM plates inside the container side wall

Fig. 13 shows a refrigerated container equipped with Phase Change Material (PCM) plates installed inside the container's left and right-side walls. The PCM plates are 30 mm thick, 300 mm wide, and 2400 mm long, extending along the container length. A total of six PCM-filled plates are installed, with three plates on each side of the container. These PCM plates aim to maintain a uniform air temperature within the container, which is crucial for preserving high-sensitivity food. The blue and red arrows on the container's top front side indicate the airflow direction. Blue arrows represent cold air entering the container, while red arrows show air exiting (outlet), which suggests the circulation mechanism to maintain the desired temperature.

4.2.2. Installed PCM-filled plate meshing

The mesh for the PCM-filled plates installed on the roof and side walls of the refrigerated container was generated using ANSYS 2020 R1.

A structured mesh employed, featuring parallel quadrilateral elements, is shown in Fig. 14. This type of mesh provides a uniform element distribution, ensuring computational accuracy and efficiency in capturing heat transfer within the PCM and its aluminum casing. The structured mesh is advantageous for the uniform geometry of the PCM-filled plates. These installations were designed to mitigate temperature fluctuations during door opening and closing, enhancing thermal regulation within the refrigerated container.

4.2.3. Numerical solver steps for phase change material (PCM) modeling

The solver settings are modified to PCM inclusion plates inside the refrigerated container. The energy model, SST $k-\omega$ model, and solidification/melting model are enabled to account for energy transfer, fluid turbulence, and PCM phase changes. The PCM material, paraffin RT8HC, is defined by properties such as density, which is assumed to be constant using the Boussinesq approximation with a value of 770 kg/m^3 , specific heat, thermal conductivity, and viscosity, as detailed in Table 5. The computational domain includes air (fluid domain), insulation (solid domain), and PCM RT8HC (phase change domain). Boundary conditions are as per Section 2.4. The coupled algorithm is used for pressure-velocity coupling, with a second-order upwind scheme for spatial discretization. The refrigerated vehicle container is initialized at 295.5 K, with PCM at 284.15 K. The time step is 1 s, with 7200 steps for door-closed and 1800 steps for door-open phases, a total of 9000 s of simulation.

4.2.4. Numerical simulation results of PCM inclusion plates inside the container side walls

Fig. 15 shows the temperature variation within the refrigerated container during two distinct phases: 120 min of closed-door operation followed by 30 min of open-door operation. The temperature profiles, shown with and without PCM-installed plates, demonstrate the PCM's impact on temperature stabilization. During the initial closed-door period, the air temperature inside the container without PCM steadily declines to approximately 0.4°C . In contrast, the container with PCM stabilizes around 0.53°C . This difference arises from the PCM's latent heat absorption during its phase change. As the temperature drops, the PCM solidifies, storing thermal energy and maintaining a more stable internal environment.

At the 7200-s mark, the door opens, introducing warmer air and turn

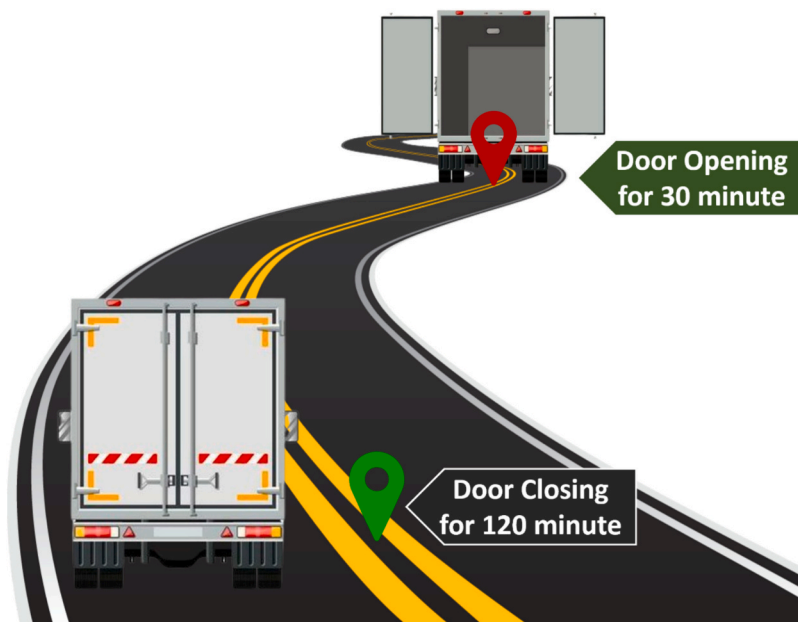


Fig. 10. Refrigerated vehicle door operation cycle.

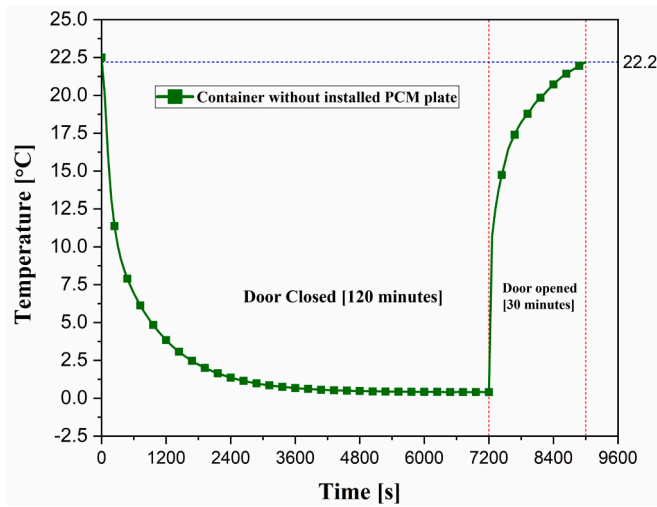


Fig. 11. Air temperature during door closed and open positions in refrigerated vehicle.

OFF the refrigeration system. The container without PCM experiences a rapid temperature increase, reaching 10.69 °C within minutes and peaking at 22.2 °C by the end of the open-door period. However, the container with PCM exhibits a slower temperature increase, peaking at 7.51 °C and reaching 18.28 °C when the door closes. The melting PCM absorbs excess heat, moderating the temperature spike and maintaining a more stable environment. This analysis shows that PCM integration reduces temperature fluctuations by approximately 4 °C during door openings, enhancing the system's thermal resilience and suitability for cold chain logistics. The PCM's dual-phase behavior ensures a consistent temperature, for preserving the quality of perishable goods.

4.2.5. Incorporation of PCM plates inside the container roof

Fig. 16 shows the arrangement of PCM (Phase Change Material) plates installed inside the roof (ceiling) of the refrigerated container. These PCM plates are intended to improve the thermal management of the container by absorbing and releasing thermal energy. Each PCM plate installed at ceiling of the container is 30 mm thick, 100 mm wide, and 2000 mm long, running along the length of the container. In total, seven plates are installed at ceiling of the container at equal distances. These plates are strategically positioned to optimize their contact with the cold air circulated by the refrigeration system. In this configuration, the PCM plates serve two primary functions. First, it comes into direct contact with the cold air from the refrigeration system, allowing it to effectively absorb and store thermal energy during the cooling phase. This stored energy is later released when the container's internal temperature begins to increase, thus aiding in keeping a stable temperature within the container. This phase-change process is essential for preventing temperature fluctuations that could compromise the quality of temperature-sensitive goods. Second, the PCM plates inside the container roof also act as airflow guides. As the cold air from the refrigeration system flows along the length of the plates, the design of the plates ensures that the air is directed evenly throughout the container. This controlled airflow helps to distribute the cooled air more uniformly, preventing hotspots and ensuring that all areas of the container are maintained at the desired temperature.

4.2.6. Numerical simulation results of PCM inclusion plates inside the container roof

Fig. 17 shows the cold air temperature profile inside a refrigerated container equipped with PCM plates installed on the roof, which are in direct contact with the cold airflow from the refrigeration system. The cold air temperature is maintained at approximately 0.58 °C with roof-mounted PCM plates and 0.53 °C with side wall-mounted plates. These

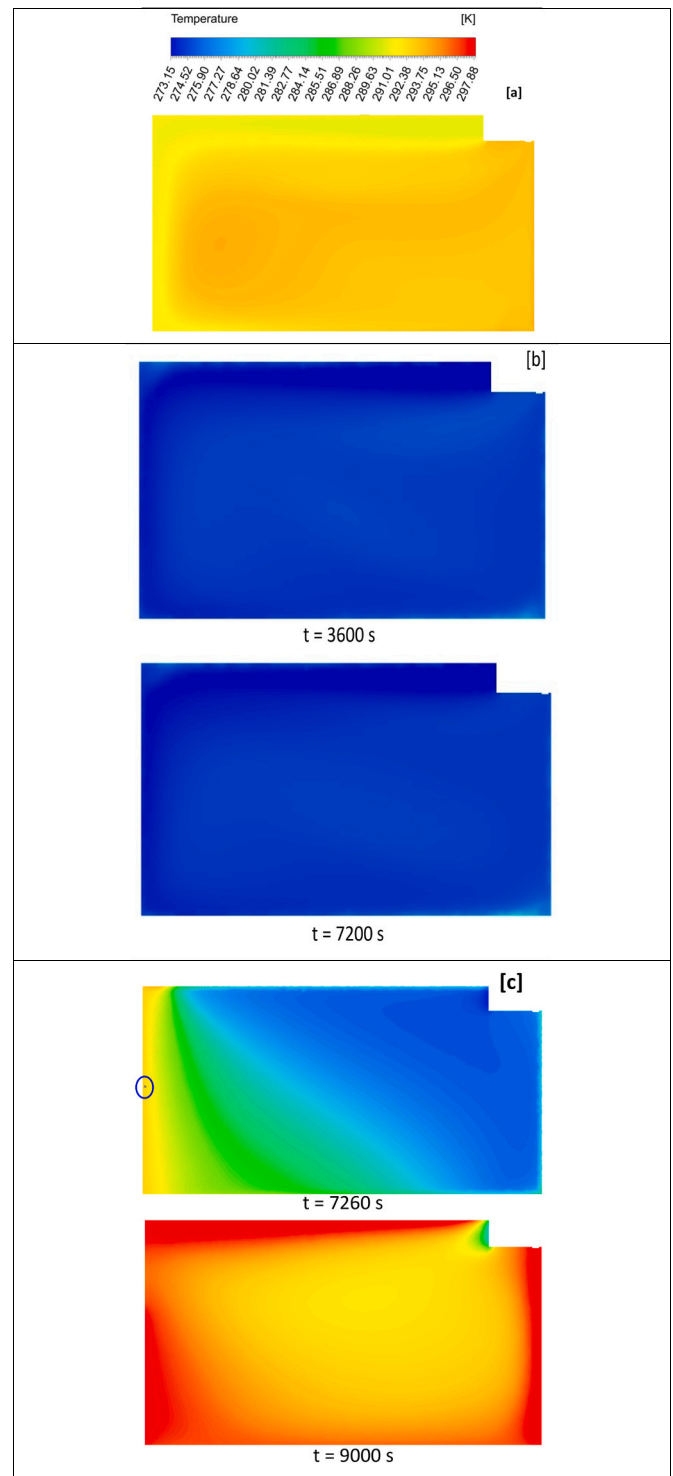


Fig. 12. Contour plot of air temperature distribution of refrigerated container.

values highlight the PCM's effectiveness in stabilizing the air temperature within a narrow range during the initial two-hour closed-door operation.

When the container door is opened, warmer air enters through natural convection, initiating a primary convective loop at the center of the container. The warm air at the bottom of this loop rises toward the roof, where it encounters the cooler PCM plates. At this point, heat is transferred from the warmer air to the PCM, which remains at a lower temperature due to its phase change properties. This process continues until the temperature gradient between the air and the PCM diminishes

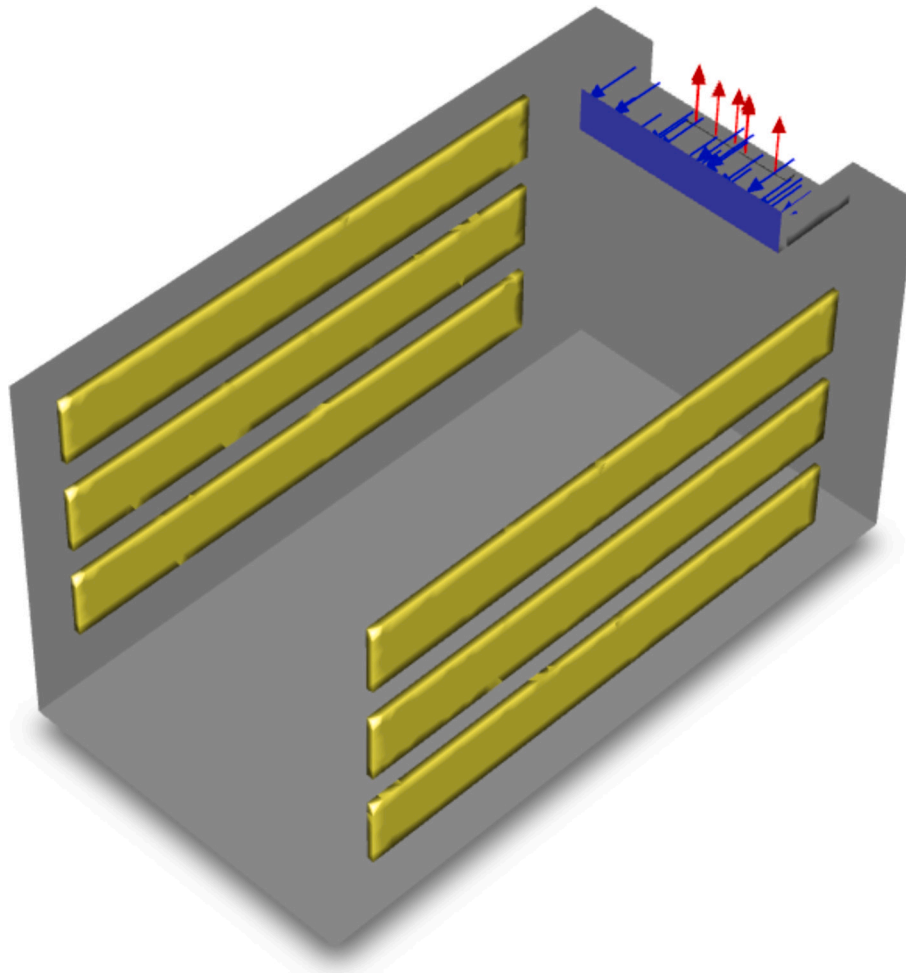


Fig. 13. 3D model of refrigerated container with incorporated PCM plates on the side walls.

significantly. The installed PCM plates effectively moderate the air temperature inside the container, reducing temperature fluctuations by $4.66\text{ }^{\circ}\text{C}$. The data in Fig. 17 further shows that the temperature stabilization performance is enhanced when PCM plates are installed on both the roof and side walls. The temperature gradient between the PCM plates on the roof and the side walls is measured at $0.74\text{ }^{\circ}\text{C}$, showcasing the system's uniform cooling capacity. These findings confirm that the PCM integration reduces temperature variability, ensuring a more stable internal environment, which is essential for preserving the quality of perishable goods during cold chain operations.

4.3. Comparative analysis of PCM performance

This Section 4.3, presents a comprehensive comparative analysis of several phase change materials (PCMs) used in refrigerated containers. The performance of a refrigerated container equipped with PCM plates is influenced by various factors, including the latent heat capacity, liquidus temperature, and solidus temperature of the PCMs which is shown in Table 5. These factors are important in determining the effectiveness of the PCM to maintain temperature stability within the container, especially during periods of door opening and closing.

Fig. 18 presents the cold air temperature profiles within a refrigerated container equipped with various PCM plates, specifically RT8HC, RT5, and RT3HC, installed on the roof and side walls. The figure highlights the comparative performance of these PCMs in mitigating air temperature fluctuations during door-opening and closed-door operations.

During the initial closed-door period, all PCM configurations

maintain the air temperature close to $0\text{ }^{\circ}\text{C}$. This stability underscores the latent heat storage capability of the PCMs, which absorb excess thermal energy, mitigating potential temperature increases. The PCMs effectively store this energy, ensuring the cold air temperature remains stable, which is an important factor in preserving temperature-sensitive goods.

When the container door is opened for 30 min, the air temperature profiles exhibit a noticeable shift, indicating the impact of the heat load introduced by the open door. Despite this, the PCM plates counteract the heat load, slowing the rate of temperature increase. The RT3HC PCM installed on the side walls limits the air temperature to approximately $18.28\text{ }^{\circ}\text{C}$ at the end of the 9000-s simulation. Notably, when the RT3HC PCM is installed on the roof, the air temperature peaks at around $17.54\text{ }^{\circ}\text{C}$, lower than the temperatures observed with the RT8HC and RT5 PCMs. These results suggest that RT3HC PCM provides superior performance in controlling temperature and reducing temperature fluctuations, particularly when installed on the roof. This finding emphasizes the importance of selecting appropriate PCM types and installation locations for optimizing thermal performance in refrigerated containers.

Fig. 19a and b shows the volume-average temperature and liquid fraction of various PCM plates (RT8HC, RT5, and RT3HC) installed on the side walls and roof of a refrigerated container. These PCMs are strategically positioned to leverage their latent heat storage capabilities, maintaining the internal air temperature near the PCM's melting point, even when the refrigeration system is turned off during door openings.

As shown in Fig. 19a, during the initial 7200 s of closed-door operation, the volume-average temperature of the PCM decreases steadily.

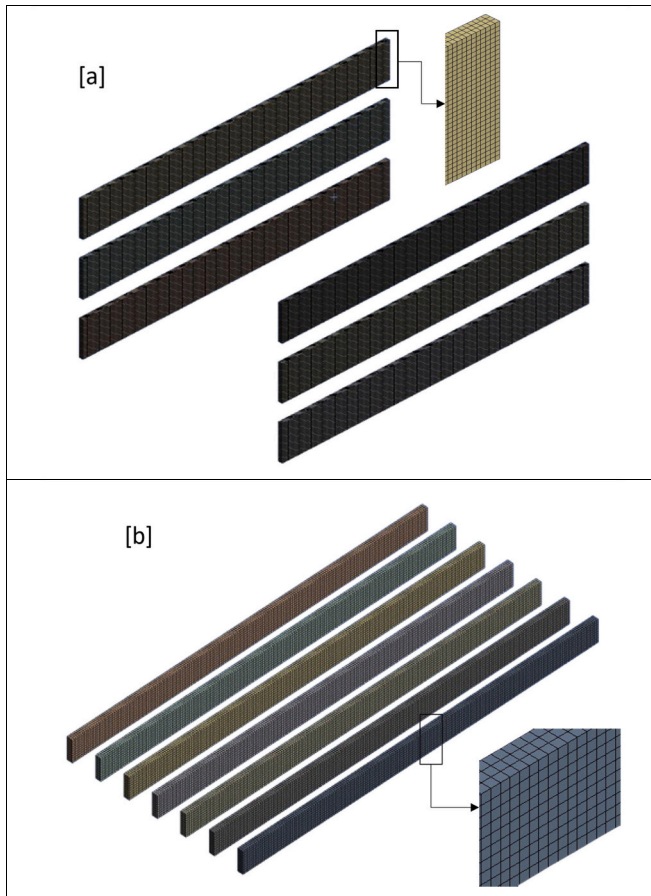


Fig. 14. Structured mesh of PCM-filled plates installed in refrigerated container. [a] Side walls. [b] Roof.

Table 5

Thermophysical properties of phase-change materials (PCMs) [32].

PCM properties			RT3HC [40]	RT5 [41]	RT8HC [42]
Melting temperature	[°C]	T_f	3	5	8
Transition temperature	[°C]	T_s	2	3	6
ΔT_{L-S}					
Density	[kg/m ³]	ρ_l	770	770	770
		ρ_s	880	880	880
Specific heat capacity	[J/kg·K]	$C_{p,l}$	2000	2000	2000
		$C_{p,s}$	2000	2000	2000
Thermal conductivity	[W/m·K]	K_l	0.2	0.2	0.2
		K_s	0.2	0.2	0.2
Latent heat of fusion	[kJ/kg]	L_f	190	180	190
Flash point	[°C]	T_{fp}	110	113	120

Among the three PCMs, only RT8HC achieves a temperature range entirely between its liquidus and solidus points, with final temperatures of 7.60 °C, 5.33 °C, and 4.56 °C for RT8HC, RT5, and RT3HC, respectively. This indicates that RT8HC is the most effective in transitioning to the solid state within this period. Fig. 19b highlights the corresponding liquid fractions, representing the proportion of PCM that remains in the liquid state. At the end of the cooling phase, the liquid fractions for RT8HC, RT5, and RT3HC are 0.8923, 0.9593, and 0.9933, respectively. This shows that RT8HC achieves the highest degree of solidification, with 10.77 % of its volume transitioning to a solid state, making it more efficient at storing latent heat.

When the door is opened for 30 min, a sharp increase in both the air temperature and the PCM's volume-average temperature is observed, as shown in Fig. 19a and b. This results from the rapid influx of warm

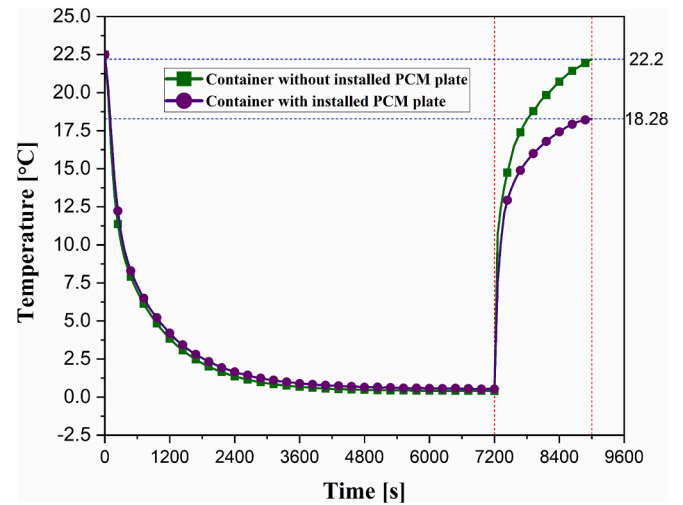


Fig. 15. Air temperature data line for refrigerated container with and without installed PCM plates.

external air, which accelerates the melting process of the PCM. The thermal load increases, and the PCM absorbs heat, causing the liquid fraction to rise. The sharp transition from solid to liquid highlights the PCM's weakened ability to buffer against the heat load during this period. The convective heat transfer mechanism is also evident during this phase. Warmer air, introduced by the open door, circulates within the container through natural convection. The cooler PCM plates on the side walls absorb heat from the rising warm air, thereby moderating the internal temperature. This heat exchange continues until a substantial temperature gradient develops between the air and the PCM, demonstrating the system's ability to partially mitigate temperature spikes caused by door openings.

Fig. 20a shows the volume-average temperature profiles of PCMs (RT3HC, RT5, and RT8HC) during the cooling phase for roof PCM plates. The temperature profiles show a gradual decrease as the PCM plates come into direct contact with the cold air from the refrigeration system. Each PCM transitions through its liquidus and solidus temperature ranges at different times due to varying thermal properties. RT8HC, having the highest thermal conductivity and latent heat, reaches its phase-change range first, at around 1620 s, and completes solidification by the end of the cooling period, attaining a liquid fraction of 0.8431. Similarly, RT5 and RT3HC enter their respective phase-change ranges at 2700 s and 3780 s, concluding the cooling process with liquid fractions of 0.8996 and 0.9510, respectively, as depicted in Fig. 20b.

It is observed that PCM plates installed on the roof achieve higher solidification efficiency compared to those installed on the side walls. This is attributed to the roof's direct exposure to circulating cold air, enhancing heat transfer, and accelerating the phase-change process. As container door is opened for 30 min, external heat infiltrates the refrigerated space, causing a rise in internal temperature. Initially, the warm air heats the lower layers near the container floor. This warm air ascends toward the roof by natural convection, resulting in a delayed thermal response at the roof level. This lag arises from the time required for the lower air layers to heat and for the thermal energy to propagate upward. During this phase, PCM plates exhibit thermal stability due to their latent heat storage capability. Fig. 20a highlights the periods of constant temperature for each PCM: approximately 4 min for RT3HC, 5 min for RT5, and 8 min for RT8HC. These intervals reflect the PCM's capacity to absorb heat without a significant rise in temperature, effectively buffering the temperature fluctuations within the refrigerated container. This performance underscores the PCM's role in mitigating thermal loads during transient operations, such as door openings, by leveraging latent heat to sustain a stable internal environment.

Fig. 21 presents a comparative analysis of RT3HC PCM plates

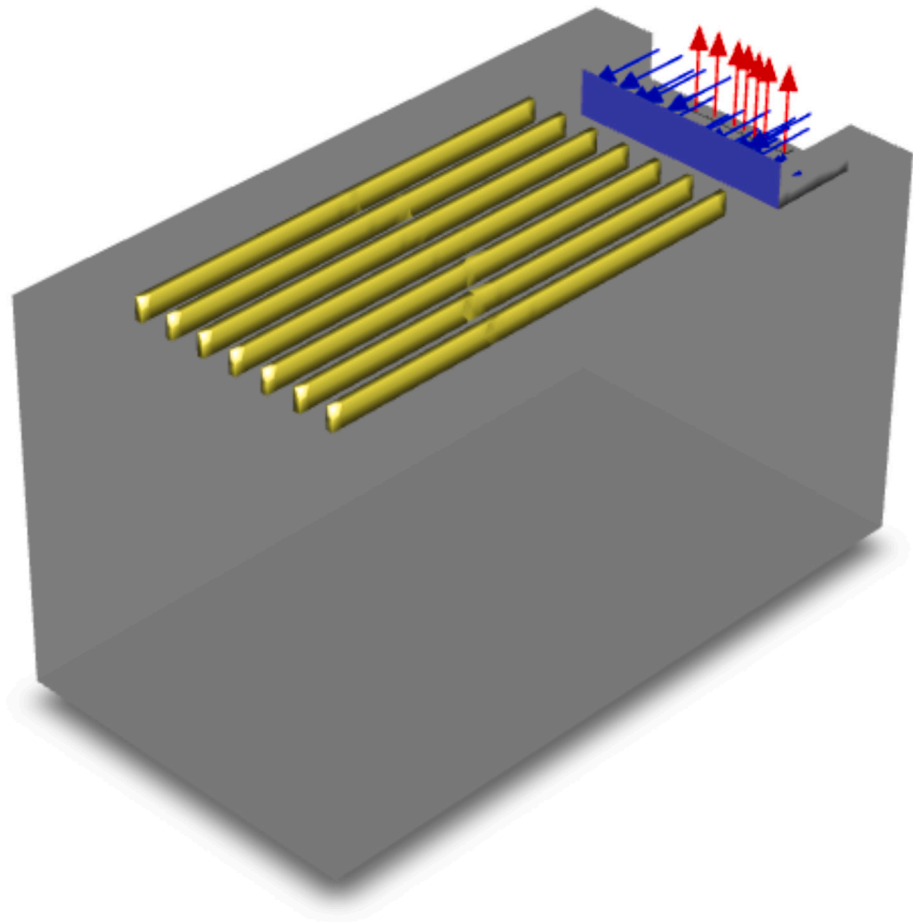


Fig. 16. 3D model of refrigerated container with incorporated PCM plates on the roof.

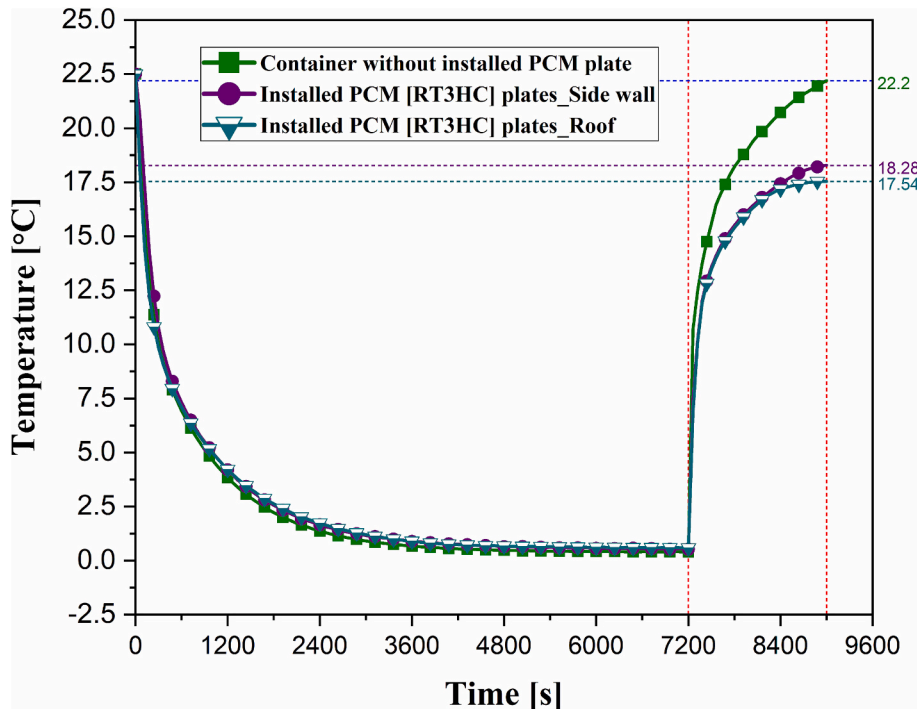


Fig. 17. Air temperature data line for roof and side wall PCM.

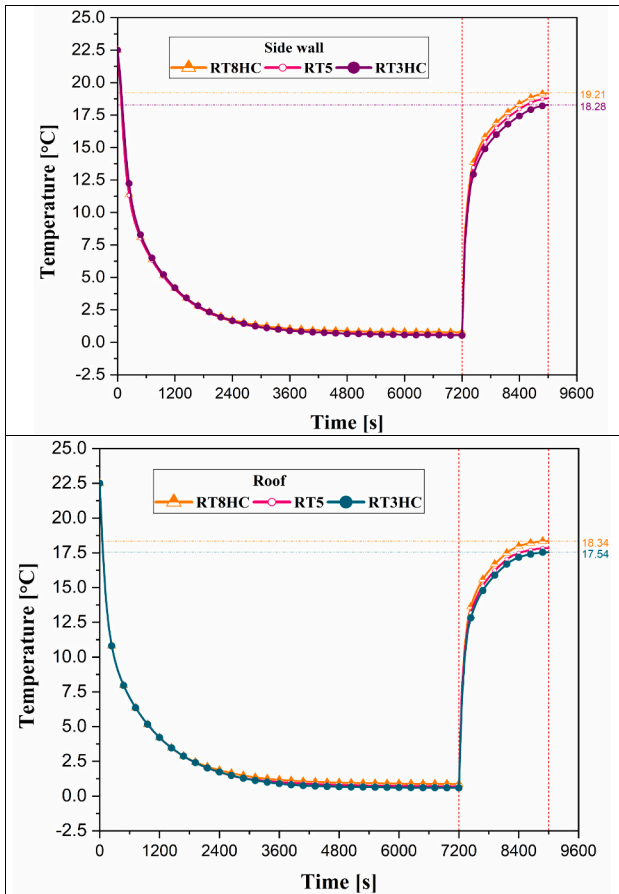


Fig. 18. Data line of cold air temperature for different PCM installed plates on the side wall and roof.

installed on the roof and side walls of a refrigerated container, focusing on their temperature profiles and solidification efficiency during the cooling process. The PCM plates on the side wall show a gradual temperature decline from 11 °C to 4.56 °C over 7200 s, maintaining a steady cooling rate due to effective heat absorption from the internal air. As the temperature approaches the PCM's liquidus point of 3 °C, a slight solidification occurs, with a liquid fraction reaching 0.9933, indicating a solidification efficiency of just 0.67 %. This implies that the PCM primarily remains in a liquid state, signifying limited solidification efficiency.

Conversely, the PCM plates installed on the roof experience faster cooling, with temperatures dropping to 3 °C within 3780 s. The liquid fraction for roof-installed PCM reaches 0.9510, reflecting a higher solidification efficiency of 4.9 %, as a more significant portion of the PCM transitions to the solid phase. This highlights the enhanced thermal energy storage capability of roof PCM plates compared to those on the side walls.

When the container door opens, warm air enters, disrupting the stable internal conditions. Both roof and side wall PCM plates absorb this external heat, causing their temperatures to rise. By the end of the 9000-s simulation, the side wall PCM temperature reaches 16.12 °C, and the roof PCM reaches 15.19 °C, with both achieving a liquid fraction of 1, signifying complete melting and maximum heat absorption. These results show the PCM's critical role in mitigating temperature fluctuations and enhancing thermal stability, with roof-installed PCM plates showing superior performance in thermal management.

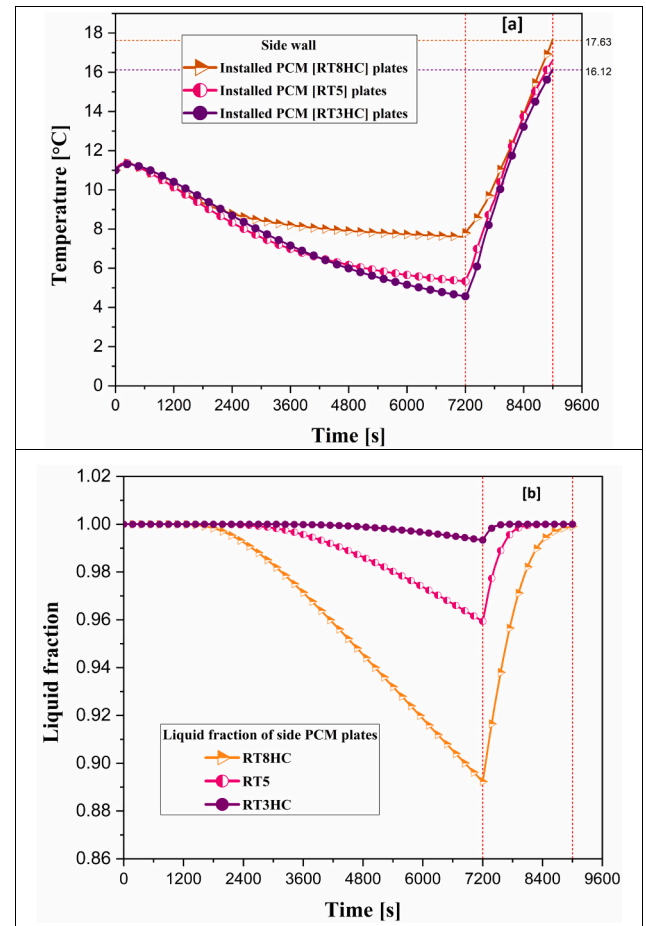


Fig. 19. (a) Variation of volume-averaged PCM temperature with time for side PCM plates.

(b) Variation of liquid fraction with time for side PCM plates.

4.4. Impact of PCM plate installation in side walls and roof on temperature distribution in refrigerated vehicles

Fig. 22 shows the arrangement of PCM-filled plates on the sides and roof of a refrigerated container. Sections 4.2.1 and 4.2.5 show the PCM plate dimensions of (30 mm × 300 mm × 2400 mm) and (30 mm × 100 mm × 2000 mm) of side plates and roof plates respectively. Six PCM-filled plates have been installed on the side walls and seven on the inside container roof. The installed PCM (Phase Change Material) plates inside the walls and on the roof of the refrigerated container are designed to observe and optimize the temperature distribution within the container. This novel arrangement aims to enhance the preservation of the quality of high-sensitivity foods by maintaining a more stable and uniform internal temperature.

Fig. 23 shows the cold air temperature variation inside a refrigerated container equipped with PCM plates of different melting temperatures, installed on both the side walls and roof. The red data line corresponds to Configuration 1, where RT8HC PCM plates, with a higher melting temperature, are installed on the side walls, while RT3HC PCM plates, with a lower melting temperature, are placed on the roof. This setup demonstrates how different PCM materials interact with cold air, effectively leveraging their thermal properties. In contrast, the blue data line represents Configuration 2, where PCM plates with a uniform melting temperature are installed on both the side walls and roof.

Both configurations exhibit similar cold air temperature trends. However, Configuration 1 maintains a lower temperature of 0.80 °C, compared to 1.04 °C in Configuration 2, indicating a better cooling

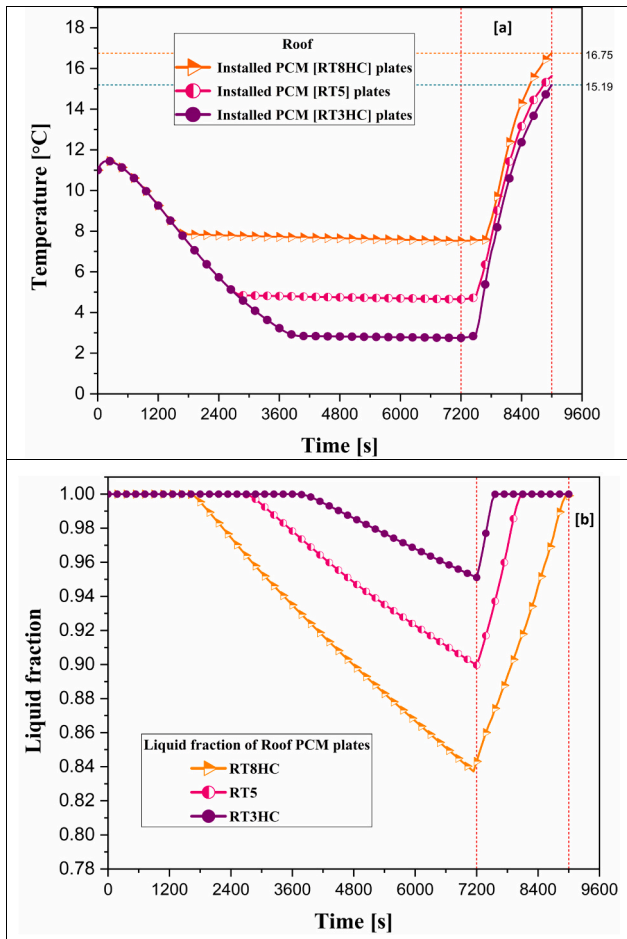


Fig. 20. (a) Variation of volume-averaged PCM temperature with time for roof PCM plates.

(b) Variation of liquid fraction with time for roof PCM plates.

performance. After the door opens for 30 min, both configurations reduce temperature fluctuations. Configuration 1 reduces the fluctuation by 5.85 °C, while Configuration 2 achieves a slightly higher reduction of 6.10 °C, compared to a container without PCM plates. This demonstrates the effectiveness of both configurations in maintaining a stable thermal environment, with Configuration 1 slightly outperforming Configuration 2 in terms of maintaining cooler air.

Fig. 24 presents a comparative analysis of two configurations in a refrigerated container. Configuration 1 uses PCM plates with varying melting temperatures: RT8HC on the side walls and RT3HC on the roof. In contrast, Configuration 2 features RT3HC PCM plates installed uniformly on both the side walls and the roof. The higher liquidus temperature of RT8HC allows it to absorb heat and release latent energy earlier in the cooling process. As the side wall RT8HC PCM plates gradually cool, they reach a temperature range between the solidus and liquidus points, initiating solidification. By the end of the cooling period, about 8.35 % of the RT8HC PCM on the side walls solidifies, with the remaining 91.65 % in liquid form. On the roof, the RT3HC PCM plates, which have a lower liquidus temperature, cool more rapidly due to direct exposure to cold air. At the end of the cooling period, only 4.76 % of the RT3HC PCM solidifies, indicating a liquid fraction of 0.9524.

In Configuration 2, RT3HC PCM plates on both the roof and side walls exhibit consistent thermal behavior. The roof PCM plates maintain a stable temperature throughout the cooling process, with a final liquid fraction of 0.95, meaning 5 % solidification. However, the side wall RT3HC PCM plates take longer to reach their liquidus temperature and show a slower solidification rate. At approximately 6720 s, these plates

begin solidifying, but only 0.67 % of the PCM transitions to a solid state within the next 8 min.

The results underscore the differing thermal behavior of PCMs with varying melting temperatures. RT8HC PCM plates on the side walls, with an initial temperature gradient of 6.64 °C, experience a more substantial cooling effect than RT3HC plates, which exhibit a smaller gradient of 2.04 °C. This temperature difference accelerates the initial heat transfer in RT8HC but slows as the gradient narrows, prolonging the solidification process. Upon opening the container door, warmer atmospheric air enters, accelerating PCM melting. Configuration 2 shows both side wall and roof RT3HC PCM plates quickly respond to the temperature increase, reaching final temperatures of 15.2 °C and 15 °C, respectively. In contrast, Configuration 1 demonstrates a slower response in the side wall RT8HC PCM due to its lower initial gradient, stabilizing at 13.7 °C after about 15 min. The dual role of PCM plates in Configuration 1—both as thermal regulators and insulating barriers—enhances the container's thermal efficiency by reducing heat ingress and stabilizing internal temperatures.

4.5. Optimized conditions for PCM integration in refrigerated containers

The numerical simulations are carried out to identify optimal configurations for PCM-filled plates to enhance temperature stability and energy efficiency in refrigerated containers. The key parameters analyzed include the placement of PCM plates and their impact on internal temperature regulation and cooling system efficiency.

Optimal plate placement: Simulations revealed that installing PCM plates along both the side walls and the ceiling yield the best thermal performance. This configuration provides uniform cooling throughout the container, significantly reducing temperature fluctuations during door opening cycles.

Performance under operating conditions: The optimized setup effectively mitigate temperature spikes during frequent door openings, reduce the temperature fluctuation by 5.85 °C to 6.10 °C in configuration 1 and configuration 2 respectively. Although the current study focused on a fixed PCM- filled thickness, future investigations could explore the impact of varying PCM layer thicknesses. Such an approach could help to identify the most cost-effective design while maintaining or enhancing thermal performance. Additionally, the influence of external factors like ambient temperature fluctuations and varying container load conditions could be further explored to optimize PCM integration.

4.6. Comparison with emerging refrigeration technologies

The PCM-based solution offers a valuable complement to advanced refrigeration systems such as thermoelectric cooling. By integrating PCMs into hybrid systems, it is possible to achieve enhanced energy efficiency and improved temperature stability. Thermoelectric cooling systems, though efficient and compact, often face challenges with temperature fluctuations due to varying thermal loads or intermittent operation. PCMs mitigate these fluctuations by leveraging their latent heat storage capacity. During periods of high thermal load, PCMs absorb excess heat through phase change (melting), preventing rapid temperature increases. Conversely, during low-load conditions or system downtime, the PCM solidifies, gradually releasing the stored heat. This thermal buffering helps to maintain a stable temperature within the cooled space. Additionally, thermoelectric modules perform within a specific temperature range. The use of PCMs helps to reduce temperature spikes, enabling the system to operate closer to its optimal efficiency range. This reduces the energy required to maintain a setpoint temperature, thus improving overall system efficiency. In applications such as refrigerated containers, the hybrid integration of PCMs with thermoelectric systems allows the effective management of peak thermal loads, leading to better temperature regulation and energy savings.

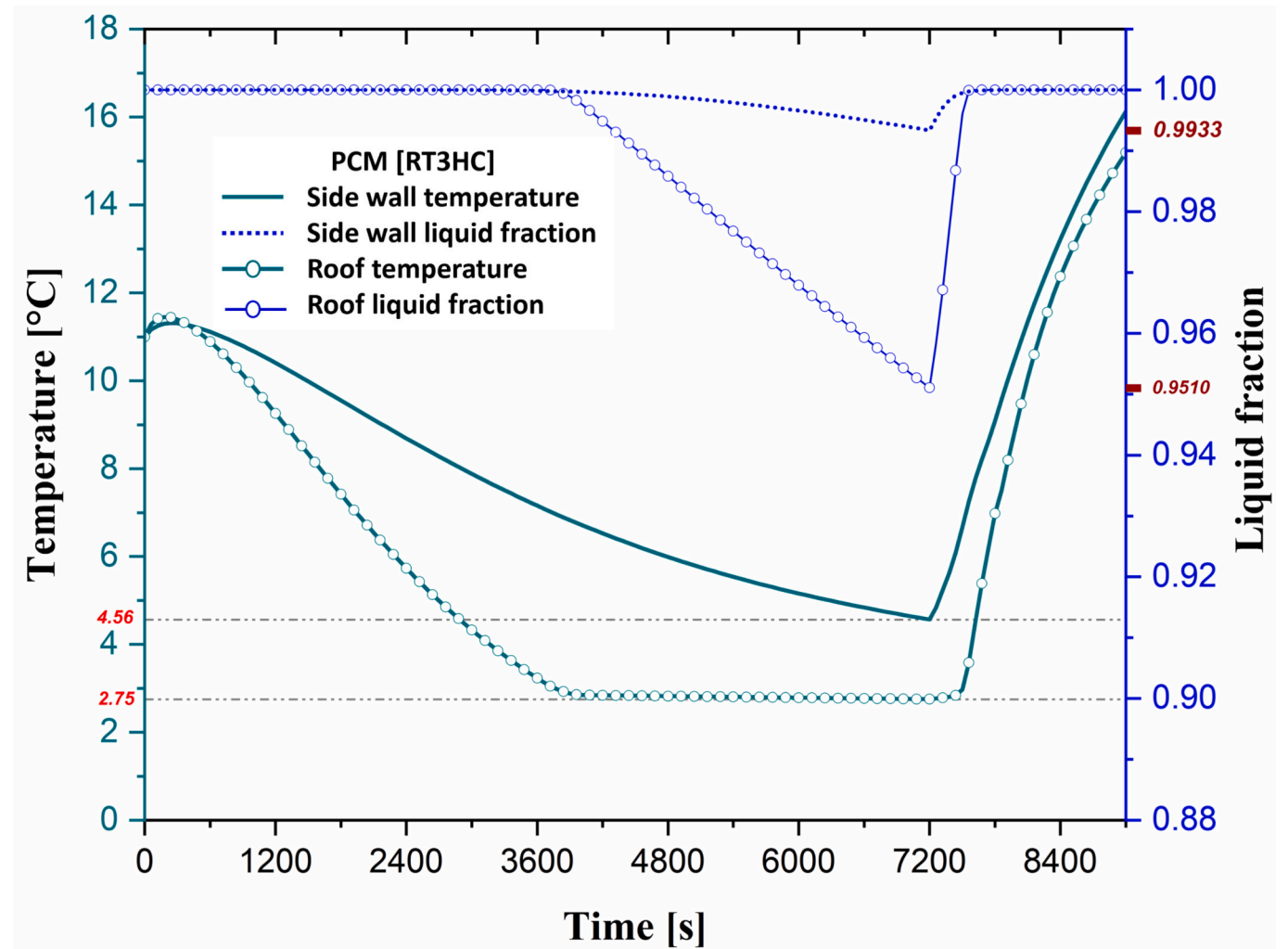


Fig. 21. Comparison of liquid fraction and PCM temperature trend, between side and roof RT3HC PCM plates.

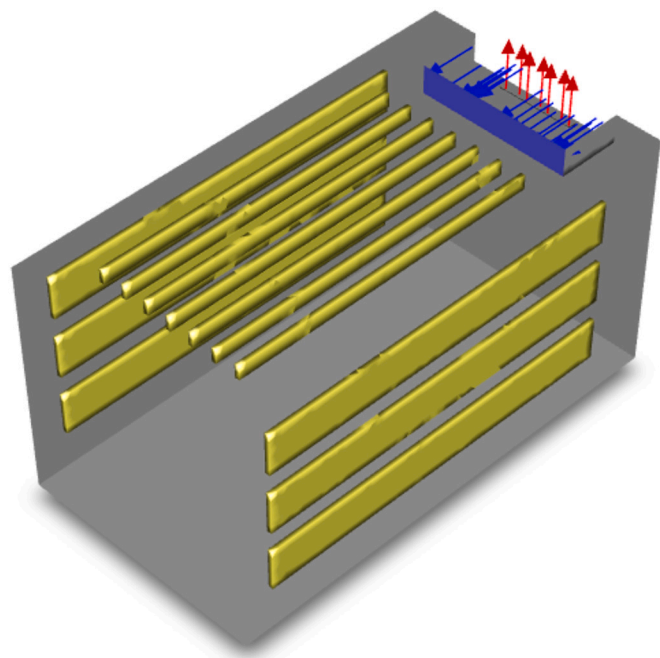


Fig. 22. 3D model of PCM-plated installed on refrigerated container.

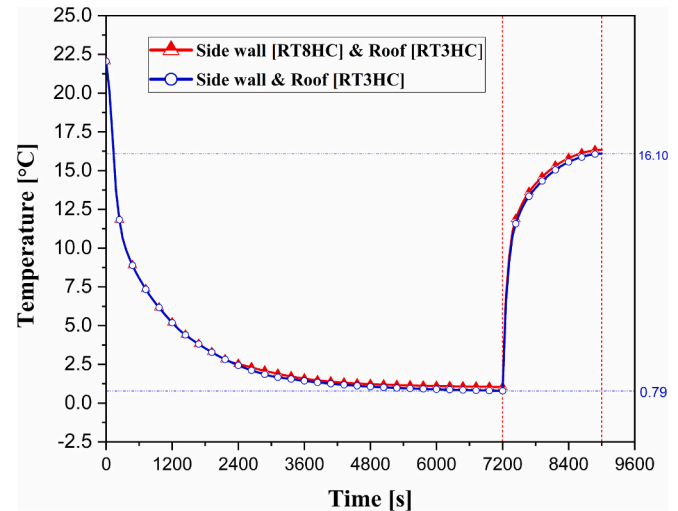


Fig. 23. Cold air temperature curve for Configuration 1 and Configuration 2.

5. Conclusion

This study investigates the performance of refrigerated container by installing innovative passive thermal energy storage plates on the side

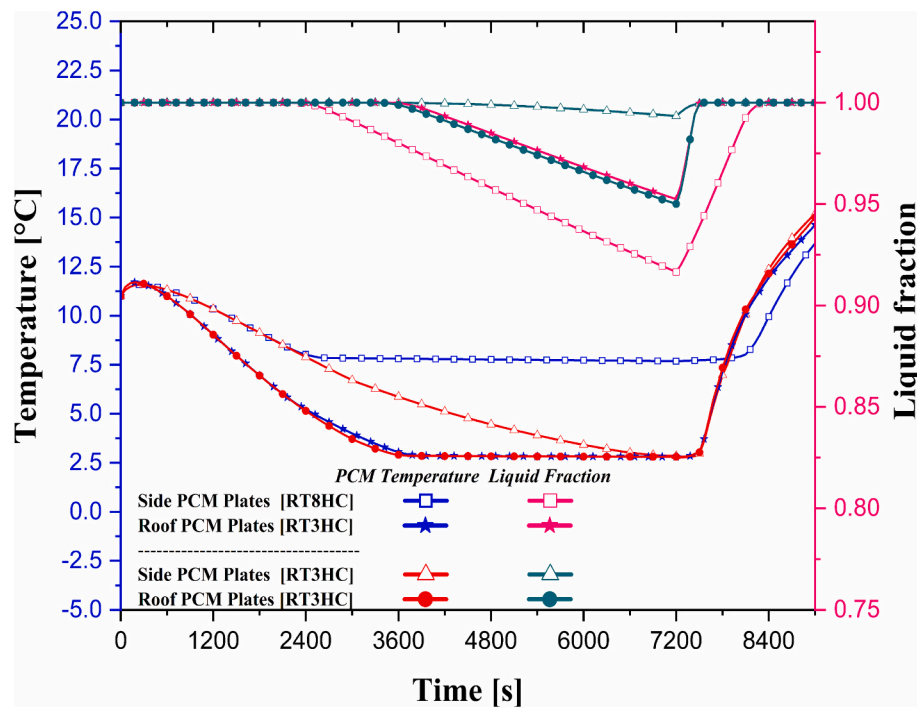


Fig. 24. Comparison of PCM temperature curve and liquid fraction between the Configuration 1 and Configuration 2.

walls and within a flow-guiding PCM channel on the ceiling of a refrigerated container. The PCM plates were carefully positioned to absorb and release thermal energy while directing airflow to enhance thermal management within the container. A detailed analysis was conducted using ANSYS Fluent to compare the effectiveness of different PCM plate configurations. The key points are summarized below:

- A passive thermal management system using PCM plates are developed to improve temperature stability inside refrigerated containers. A novel approach of strategically positioning of PCM plates is implemented to manage temperature fluctuations efficiently during simulated real-world conditions, such as door openings.
- An ANSYS Fluent is used to simulate temperature distribution and phase change behavior. Three paraffin-based PCMs (RT3HC, RT5, and RT8HC) with varying melting temperatures were tested. Temperature fluctuations and solidification efficiency were analyzed under door-opening scenarios.
- **Effectiveness of PCM Plates:** Installing PCM plates on the ceiling and side walls of the refrigerated container significantly reduces air temperature fluctuations, achieving a reduction of 4.66 °C and 3.92 °C, respectively. Using RT8HC PCM plates on the side walls shows increase in a solidification efficiency by 10.77 % than RT3HC, giving to better thermal stability.
- **Optimization of PCM Configuration:** When PCM-filled plates are installed on both the side walls and roof, the temperature fluctuation is minimized. Further, cold air fluctuation reduced by 5.85 °C and 6.10 °C in two configurations. Configuration 1, with earlier liquidus temperature, provides higher efficiency, enhancing the overall performance of the thermal energy storage system
- The current study use 30 mm thick PCM plates on the side walls, serving dual roles in thermal regulation and insulation. Future research will investigate varying PCM plate thicknesses (20 mm, 25 mm, 40 mm) to optimize thermal performance further. Additional studies will incorporate factors such as airflow and heat loss through the refrigeration equipment for comprehensive modeling.

CRediT authorship contribution statement

Tushar B. Umate: Writing – original draft, Validation, Investigation, Data curation, Conceptualization. **Pravin D. Sawarkar:** Writing – review & editing, Supervision, Methodology.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors used ChatGPT 3.5 to improve the paper's readability and language. After using this tool/service, the authors reviewed and edited the content as necessary, and they accept full responsibility for the publication's content.

Declaration of competing interest

The authors confirm that they do not have any apparent competing financial interests or personal relationships that could have affected the integrity of the work presented in this paper.

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Data availability

Data will be made available on request.

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